Are elevated lactate levels associated with ICU mortality?

...

January, 2025

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Authors' information

Nombre	Johanna Ursula Albers
Email	johanna.albers@estudiants.urv.cat
GitHub	https://github.com/
LinkedIn	https://www.linkedin.com/in/
Fecha	January 23, 2026

Nombre	Marco Russo
Email	mrussorb@uoc.edu
GitHub	https://github.com/marcusRB/
LinkedIn	https://www.linkedin.com/in/marcusrb/
Fecha	January 23, 2026

Nombre	Noa-Sibila Janer Oliver
Email	noa-sibila.janer@estudiants.urv.cat
GitHub	https://github.com/
LinkedIn	https://www.linkedin.com/in/
Fecha	January 23, 2026

1 Project overview

Research Question: Are elevated lactate levels associated with ICU mortality? Study Design: Retrospective cohort study using MIMIC-III database Primary Analysis: Association between first lactate measurement post-ICU admission and in-hospital mortality.

1.1 Data Description

MIMIC-III is a relational database consisting of 26 tables. Tables are linked by identifiers which usually have the suffix 'ID'. For example, SUBJECT_ID refers to a unique patient, HADM_ID refers to a unique admission to the hospital, and ICUSTAY_ID refers to a unique admission to an intensive care unit.

Charted events such as notes, laboratory tests, and fluid balance are stored in a series of 'events' tables. For example the OUTPUTEVENTS table contains all measurements related to output for a given patient, while the LABEVENTS table contains laboratory test results for a patient.

Tables prefixed with 'D_' are dictionary tables and provide definitions for identifiers. For example, every row of CHARTEVENTS is associated with a single ITEMID which represents the concept measured, but it does not contain the actual name of the measurement. By joining CHARTEVENTS and D_ITEMS on ITEMID, it is possible to identify the concept represented by a given ITEMID.

Developing the MIMIC data model involved balancing simplicity of interpretation against closeness to ground truth. As such, the model is a reflection of underlying data sources, modified over iterations of the MIMIC database in response to user feedback. Care has been taken to avoid making assumptions about the underlying data when carrying out transformations, so MIMIC-III closely represents the raw hospital data.

Broadly speaking, five tables are used to define and track patient stays: ADMISSIONS; PATIENTS; ICUSTAYS; SERVICES; and TRANSFERS. Another five tables are dictionaries for cross-referencing codes against their respective definitions: D_CPT; D_ICD_DIAGNOSES; D_ICD_PROCEDURES; D_ITEMS; and D_LABITEMS. The remaining tables contain data associated with patient care, such as physiological measurements, caregiver observations, and billing information.

In some cases it would be possible to merge tables—for example, the D_ICD_PROCEDURES and CPTEVENTS tables both contain detail relating to procedures and could be combined—but our approach is to keep the tables independent for clarity, since the data sources are significantly different. Rather than combining the tables within MIMIC data model, we suggest researchers develop database views and transforms as appropriate.

1.2 MIMIC-III v1.4

The current version of the database is v1.4. When referencing this version, we recommend using the full title: MIMIC-III v1.4[1].

MIMIC-III v1.4 was released on 2 September 2016. It was a major release enhancing data quality and providing a large amount of additional data for Metavision patients.

---

[1] Johnson, A., Pollard, T., & Mark, R. (2016). MIMIC-III Clinical Database (version 1.4). PhysioNet. RRID:SCR_007345. https://doi.org/10.13026/C2XW26

2 Data Extraction & Preparation

2.1 Database setup, cohort extraction

```r
# Creating the main directories
if (!dir.exists("../figures")) {
dir.create("../figures", recursive = TRUE)
}

if (!dir.exists("../results")) {
dir.create("../results", recursive = TRUE)
}
```

We constructed a cohort at the level of individual ICU stays. Each row in the dataset corresponds to a single ICU stay. The most important measurement, the first lactate measurement after the ICU admission is included.

Lactate measurements were assigned to ICU stays based on their timestamp: a measurement was linked to an ICU stay only if it occurred between the ICU admission (intime) and 24h after.

ICU mortality was derived from hospital admission outcomes. For patients with multiple ICU stays within the same hospital admission, ICU mortality was assigned exclusively to the final ICU stay if the patient died during that admission; all preceding ICU stays were labeled as non-fatal. This approach allows ICU-level mortality attribution in the absence of a direct ICU death indicator. The final cohort comprised 129,966 lactate measurements, representing multiple ICU stays and hospital admissions across patients.

Further confounding variables are also included: age, gender, creatine kinase, bilirubin and mean arterial blood pressure (map). The inlcuded measurements are the first measurements in the first 24 h after the ICU admission.

```r
initial_df <- dbGetQuery(con, cohort_query)
initial_df %>% glimpse()
```

```
## Rows: 22,307
## Columns: 31
## $ row_id             <int64> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 1~
## $ subject_id         <int> 3, 9, 12, 21, 25, 31, 33, 36, 36, 38, 41, 41, 45, ~
## $ dob                <dttm> 2025-04-11, 2108-01-26, 2032-03-24, 2047-04-04, 2~
## $ age_icu_entry      <int> 76, 41, 72, 87, 59, 72, 82, 70, 73, 76, 57, 57, 42~
## $ age_icu_exit       <int> 76, 41, 72, 87, 59, 72, 82, 70, 73, 76, 57, 57, 42~
## $ gender             <chr> "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", ~
## $ hadm_id            <int> 145834, 150750, 112213, 109451, 129635, 128652, 17~
## $ icustay_id         <int> 211552, 220597, 232669, 217847, 203487, 254478, 29~
## $ intime             <dttm> 2101-10-20 19:10:11, 2149-11-09 13:07:02, 2104-08~
## $ outtime            <dttm> 2101-10-26 20:43:09, 2149-11-14 20:52:14, 2104-08~
## $ hospital_dead      <int> 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ icu_dead           <int> 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ lactate_time       <dttm> 2101-10-20 19:12:00, 2149-11-09 17:47:00, 2104-08~
## $ lactate_value      <chr> "4.3", "1.9", "13.8", "1.9", "1.6", "1.4", "6.0", ~
## $ lactate_value_num  <dbl> 4.3, 1.9, 13.8, 1.9, 1.6, 1.4, 6.0, 6.1, 1.0, 2.8,~
## $ lactate_units      <chr> "mmol/L", "mmol/L", "mmol/L", "mmol/L", "mmol/L", ~
## $ lactate_flag       <chr> "abnormal", NA, "abnormal", NA, NA, NA, "abnormal"~
## $ ck_time            <dttm> 2101-10-20 19:59:00, NA, 2104-08-08 07:30:00, 213~
```

```
## $ ck_value            <chr> "82", NA, "1554", "593", "296", NA, NA, NA, "347",~
## $ ck_value_num        <dbl> 82, NA, 1554, 593, 296, NA, NA, NA, 347, NA, NA, N~
## $ ck_units            <chr> "IU/L", NA, "IU/L", "IU/L", "IU/L", NA, NA, NA, "I~
## $ ck_flag             <chr> NA, NA, "abnormal", "abnormal", "abnormal", NA, NA~
## $ bilirubin_time      <dttm> 2101-10-20 19:59:00, 2149-11-10 09:40:00, NA, 213~
## $ bilirubin_value     <chr> "0.8", "0.4", NA, "0.4", "0.4", NA, NA, "0.4", NA,~
## $ bilirubin_value_num <dbl> 0.8, 0.4, NA, 0.4, 0.4, NA, NA, 0.4, NA, NA, NA, N~
## $ bilirubin_units     <chr> "mg/dL", "mg/dL", NA, "mg/dL", "mg/dL", NA, NA, "m~
## $ bilirubin_flag      <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ mabp_time           <dttm> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ mabp_value          <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ mabp_value_num      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ mabp_units          <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
```

2.2 Data cleaning

First we checked the data for duplicated. No duplicates were found.

```r
# Are there duplicates?
sum(duplicated(initial_df))
```

```
## [1] 0
```

Checking the initial dataframe, we saw some null values to analyse if to impute or to remove in the next steps.

```r
as.data.frame(initial_df) %>%
  summarise(across(everything(), ~ sum(is.na(.)))) %>%
  select(where(~ . > 0))
```

```
##   lactate_value_num lactate_flag ck_time ck_value ck_value_num ck_units ck_flag
## 1                 5        12772   13113    13113        13114    13113   17411
##   bilirubin_time bilirubin_value bilirubin_value_num bilirubin_units
## 1          12615           12615               12616           12615
##   bilirubin_flag mabp_time mabp_value mabp_value_num mabp_units
## 1          19575     15772      15772          15772      15772
```

We only kept observations, which are greater then 0 mmol/L, made sure that all lactate values are numeric and checked for suspicious values.

```r
clean_values_df <- initial_df %>%
  mutate(
    lactate_value_num = case_when(
      # For samples containing ">" or "GREATER THAN 30" assign value 31

      str_detect(lactate_value, fixed(">")) |
      str_detect(toupper(lactate_value), "GREATER THAN") ~ 31,

      TRUE ~ as.numeric(lactate_value) # Convert to numeric value,
```

```
      # samples with numbers and letters will be converted to NA
    )
  ) %>%
  filter(!is.na(lactate_value_num)) # Delete whole row if lactate value is null


# Check for suspicious values
clean_values_df %>%
  filter(lactate_value_num <= 0 | lactate_value_num > 31)
```

```
##  [1] row_id              subject_id          dob
##  [4] age_icu_entry       age_icu_exit        gender
##  [7] hadm_id             icustay_id          intime
## [10] outtime             hospital_dead       icu_dead
## [13] lactate_time        lactate_value       lactate_value_num
## [16] lactate_units       lactate_flag        ck_time
## [19] ck_value            ck_value_num        ck_units
## [22] ck_flag             bilirubin_time      bilirubin_value
## [25] bilirubin_value_num bilirubin_units     bilirubin_flag
## [28] mabp_time           mabp_value          mabp_value_num
## [31] mabp_units
## <0 rows> (or 0-length row.names)
```

We verified the logical consistency of mortality indicators by confirming that ICU death did not occur in the absence of in-hospital death.

```
clean_values_df %>%
  filter(icu_dead == 1 & hospital_dead == 0) %>%
  nrow()
```

```
## [1] 0
```

We assessed mortality consistency at the patient level by confirming that each subject had at most one recorded ICU death across all ICU stays.

```
death_counts <- clean_values_df %>%
  group_by(subject_id) %>%
  summarise(
    hosp_dead = n_distinct(hadm_id[hospital_dead == 1], na.rm = TRUE),
    icu_dead  = n_distinct(icustay_id[icu_dead == 1], na.rm = TRUE)
  ) %>%
  filter(icu_dead > 1)
death_counts
```

```
## # A tibble: 1 x 3
##   subject_id hosp_dead icu_dead
##        <int>     <int>    <int>
## 1      17796         2        2
```

A data inconsistency was identified involving a single subject with two incorrectly assigned mortality flags; these records were manually rectified to ensure data integrity.

```
clean_values_df <- clean_values_df %>%
  mutate(
    # Reassign hospital_dead from first admission
    hospital_dead = if_else(subject_id == 17796 & hadm_id == 119823, 0, hospital_dead),

    # Reassign icu_dead from first ICU stay
    icu_dead = if_else(subject_id == 17796 & hadm_id == 119823, 0, icu_dead)
  )
```

For our analysis, we will only need the following columns: `icustay_id`, `age_icu_exit`, `gender`, `icu_dead`, `lactate_value_num`, `ck_value_num`, `bilirubin_value_num`, `mabp_value_num`. Regarding the age, we chose the attribute exiting the ICU, because it is the age of death or survival. These columns will be defined in *analysis_df*.

Laboratory abnormality flags were excluded from analysis, as continuous laboratory measurements provide greater informational content and allow more precise modeling than qualitative classifications.

```
clean_values_df <- clean_values_df %>%
  select(
    icustay_id, age_icu_exit, gender, icu_dead, lactate_value_num, ck_value_num, bilirubin_value_num, ma
  )

head(clean_values_df)
```

```
##   icustay_id age_icu_exit gender icu_dead lactate_value_num ck_value_num
## 1     211552           76      M        0               4.3           82
## 2     220597           41      M        1               1.9           NA
## 3     232669           72      M        1              13.8         1554
## 4     217847           87      M        0               1.9          593
## 5     203487           59      M        0               1.6          296
## 6     254478           72      M        1               1.4           NA
##   bilirubin_value_num mabp_value_num
## 1                 0.8             NA
## 2                 0.4             NA
## 3                  NA             NA
## 4                 0.4             NA
## 5                 0.4             NA
## 6                  NA             NA
```

```
summary(clean_values_df)
```

```
##    icustay_id      age_icu_exit       gender             icu_dead
##  Min.   :200001   Min.   :  0.00   Length:22302       Min.   :0.0000
##  1st Qu.:224940   1st Qu.: 54.00   Class :character   1st Qu.:0.0000
##  Median :249918   Median : 66.00   Mode  :character   Median :0.0000
##  Mean   :249906   Mean   : 74.09                      Mean   :0.1598
##  3rd Qu.:275104   3rd Qu.: 78.00                      3rd Qu.:0.0000
##  Max.   :299993   Max.   :311.00                      Max.   :1.0000
##
##  lactate_value_num  ck_value_num    bilirubin_value_num mabp_value_num
##  Min.   : 0.300    Min.   :     3   Min.   : 0.000      Min.   :-38.00
##  1st Qu.: 1.200    1st Qu.:    60   1st Qu.: 0.400      1st Qu.: 66.00
##  Median : 1.800    Median :   140   Median : 0.800      Median : 78.00
```

```
##  Mean   : 2.436    Mean   :  1059   Mean   : 2.166      Mean   : 77.86
##  3rd Qu.: 2.900    3rd Qu.:   433   3rd Qu.: 1.800      3rd Qu.: 90.00
##  Max.   :27.000    Max.   :266720   Max.   :79.000      Max.   :361.00
##                    NA's   :13112    NA's   :12614       NA's   :15767
```

After the first step of the data cleaning, reducing the variables in 8 from 31.

Creatine kinase (CK) and bilirubin values were inspected for plausibility. Although both variables showed highly skewed distributions and substantial missingness, extreme values were considered clinically plausible (e.g., severe rhabdomyolysis for CK and advanced liver failure for bilirubin). Therefore, no values were removed during data cleaning.

Due to de-identification procedures in MIMIC-III, patients aged 89 years or older have artificially shifted dates of birth. Therefore me have a maximum of 311 in the column age_icu_exit. Age values exceeding 89 years were therefore capped at 90 years for analysis.

```
clean_values_df <- clean_values_df %>%
  mutate(
    age_icu_exit = if_else(age_icu_exit > 89, 90, age_icu_exit)
  )
```

Physiologically implausible mean arterial blood pressure values (<30 or >200 mmHg) were set to missing.

This will be our final step before the EDA, so we will simplify the name of the data frame to `eda_df`.

```
eda_df <- clean_values_df %>%
  mutate(
    mabp_value_num = if_else(
      mabp_value_num < 30 | mabp_value_num > 200,
      NA_real_,
      mabp_value_num
    )
  )
```

2.3 Exploratory analysis

We start showing the shape and short summary of the dataframe.

```
dim(eda_df)
```

```
## [1] 22302     8
```

```
summary(eda_df)
```

```
##    icustay_id      age_icu_exit      gender             icu_dead
##  Min.   :200001   Min.   : 0.00   Length:22302       Min.   :0.0000
##  1st Qu.:224940   1st Qu.:54.00   Class :character   1st Qu.:0.0000
##  Median :249918   Median :66.00   Mode  :character   Median :0.0000
##  Mean   :249906   Mean   :64.27                      Mean   :0.1598
##  3rd Qu.:275104   3rd Qu.:78.00                      3rd Qu.:0.0000
##  Max.   :299993   Max.   :90.00                      Max.   :1.0000
##
##  lactate_value_num  ck_value_num    bilirubin_value_num mabp_value_num
##  Min.   : 0.300    Min.   :     3   Min.   : 0.000      Min.   : 30.00
##  1st Qu.: 1.200    1st Qu.:    60   1st Qu.: 0.400      1st Qu.: 69.00
##  Median : 1.800    Median :   140   Median : 0.800      Median : 79.00
##  Mean   : 2.436    Mean   :  1059   Mean   : 2.166      Mean   : 80.92
##  3rd Qu.: 2.900    3rd Qu.:   433   3rd Qu.: 1.800      3rd Qu.: 90.00
##  Max.   :27.000    Max.   :266720   Max.   :79.000      Max.   :199.00
##                    NA's   :13112    NA's   :12614       NA's   :16425
```

The data frame *comprises 22,302 ICU stays*, including lactate observations, gender and age, providing a robust longitudinal perspective on patient biomarkers.

We describe which patients are in our data set.

```
table(eda_df$gender)
```

```
##
##     F     M
##  9496 12806
```

```
table(clean_values_df$icu_dead)
```

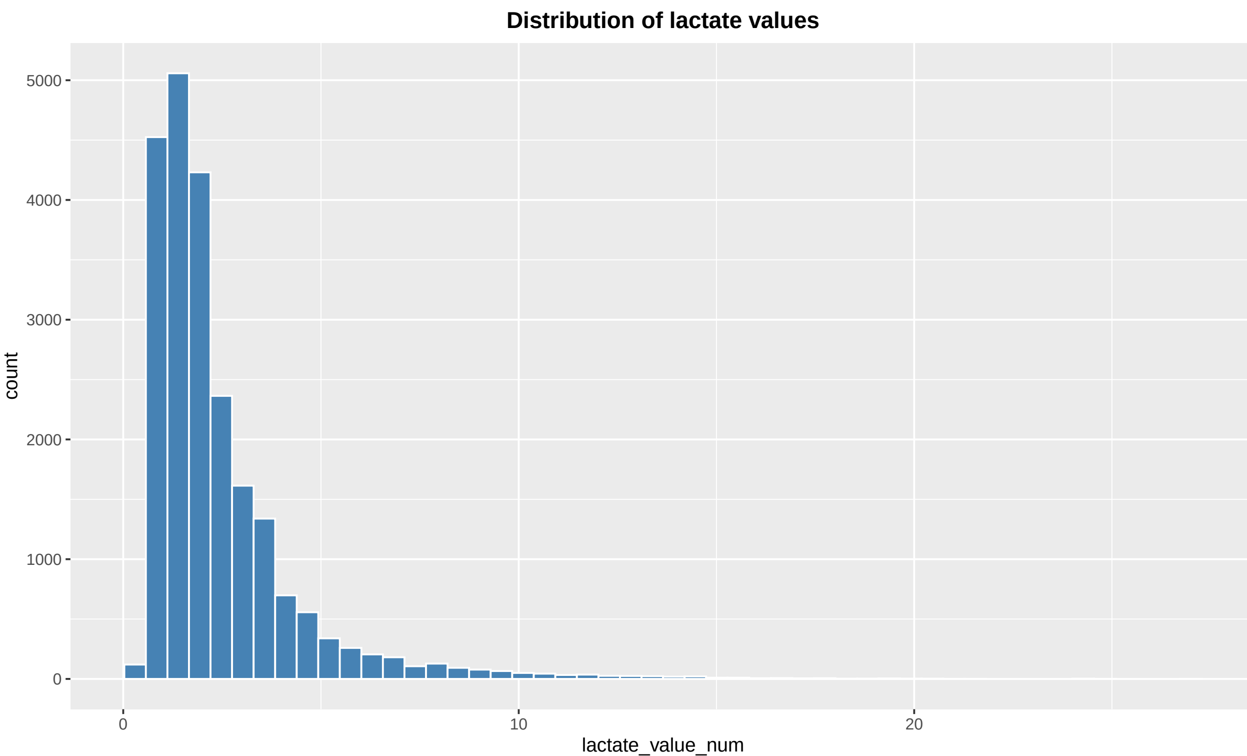
```
##
##     0     1
## 18738  3564
```

```
prop.table(table(clean_values_df$icu_dead))*100
```

```
##
##        0        1
## 84.01937 15.98063
```

We analyse the distribution of the lactate value.

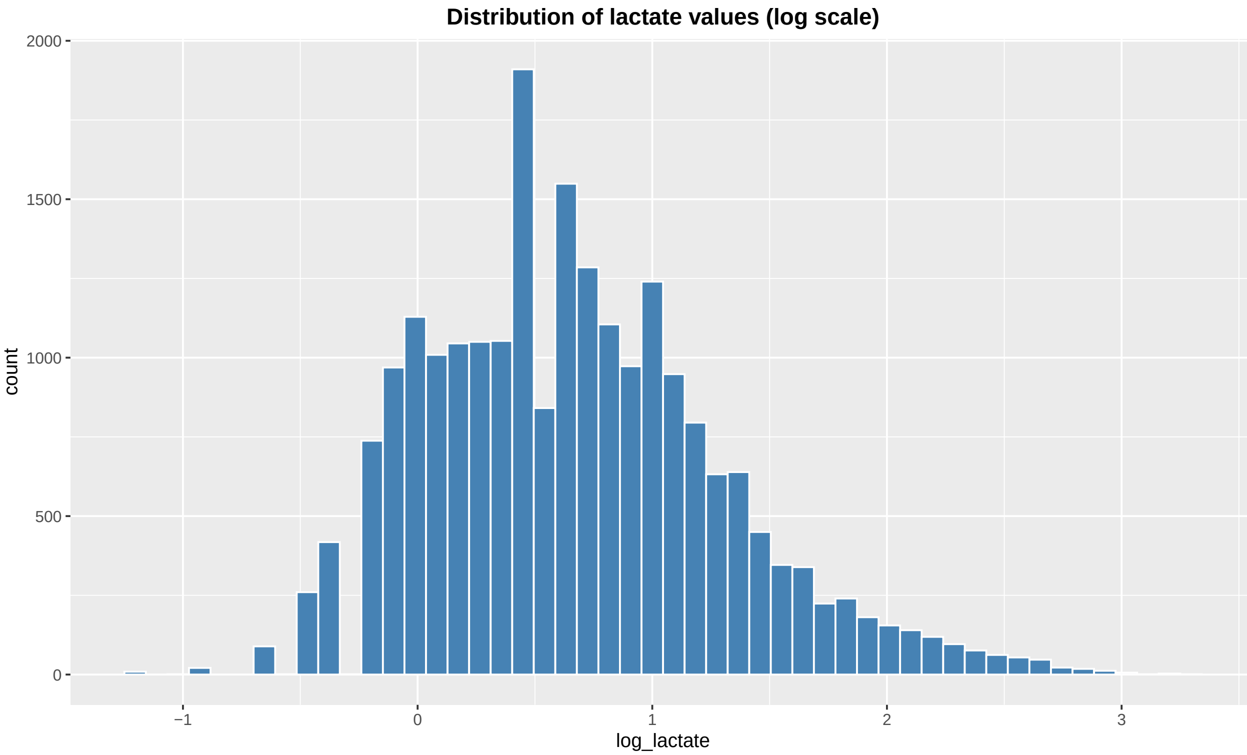
```
# Show the distribution of lactate values
figure_1 <- ggplot(eda_df, aes(lactate_value_num)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  labs(title = "Distribution of lactate values") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
figure_1
```



We add a column with the log-transformation of the lactate measurement because the values are skewed.

```
eda_df <- eda_df %>%
  mutate(
    log_lactate = log(lactate_value_num)
  )
```

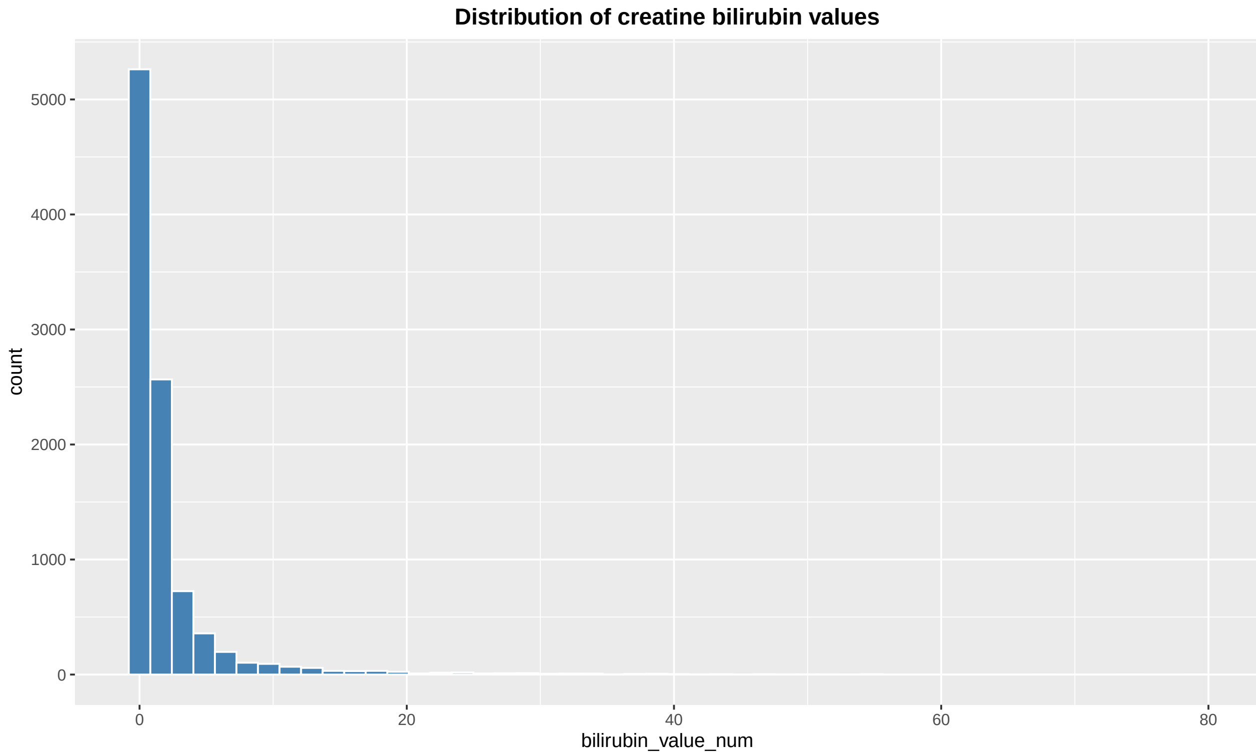
```
# Show the distribution of lactate values (log scale)
figure_2 <- ggplot(eda_df, aes(log_lactate)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  labs(title = "Distribution of lactate values (log scale)") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
figure_2
```




```
ggsave(
  filename = "../figures/figure02_distribution_lactate_value_log.png",
  plot = figure_2,
  width = 15,
  height = 5,
  units = "in",
  dpi = 300
)
```

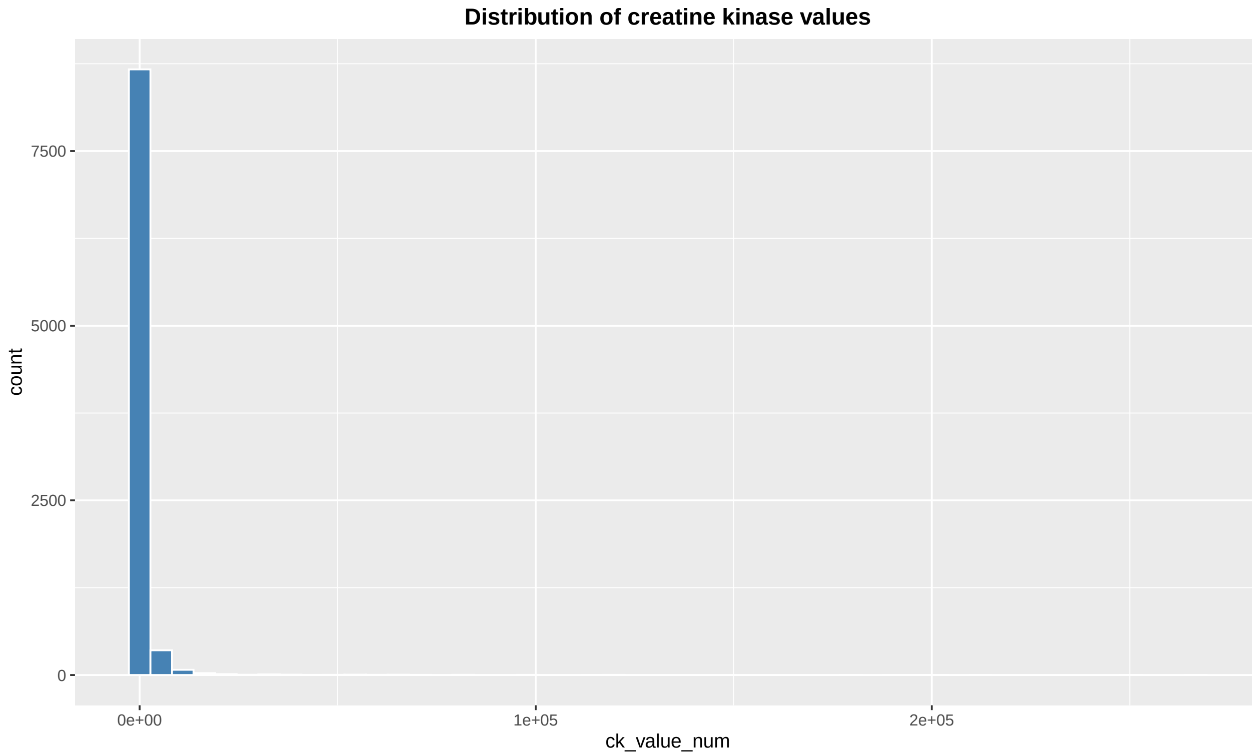
We analyse the distribution of confounding variables.

```
# Show the distribution of bilirubin values
figure_3 <- ggplot(eda_df, aes(bilirubin_value_num)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  labs(title = "Distribution of creatine bilirubin values") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

figure_3
```




```
ggsave(
  filename = "../figures/figure03_distribution_bilirubin_value.png",
  plot = figure_3,
  width = 15,
  height = 5,
  units = "in",
  dpi = 300
)
```

```
# Show the distribution of ck values
figure_4 <- ggplot(eda_df, aes(ck_value_num)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  labs(title = "Distribution of creatine kinase values") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
figure_4
```

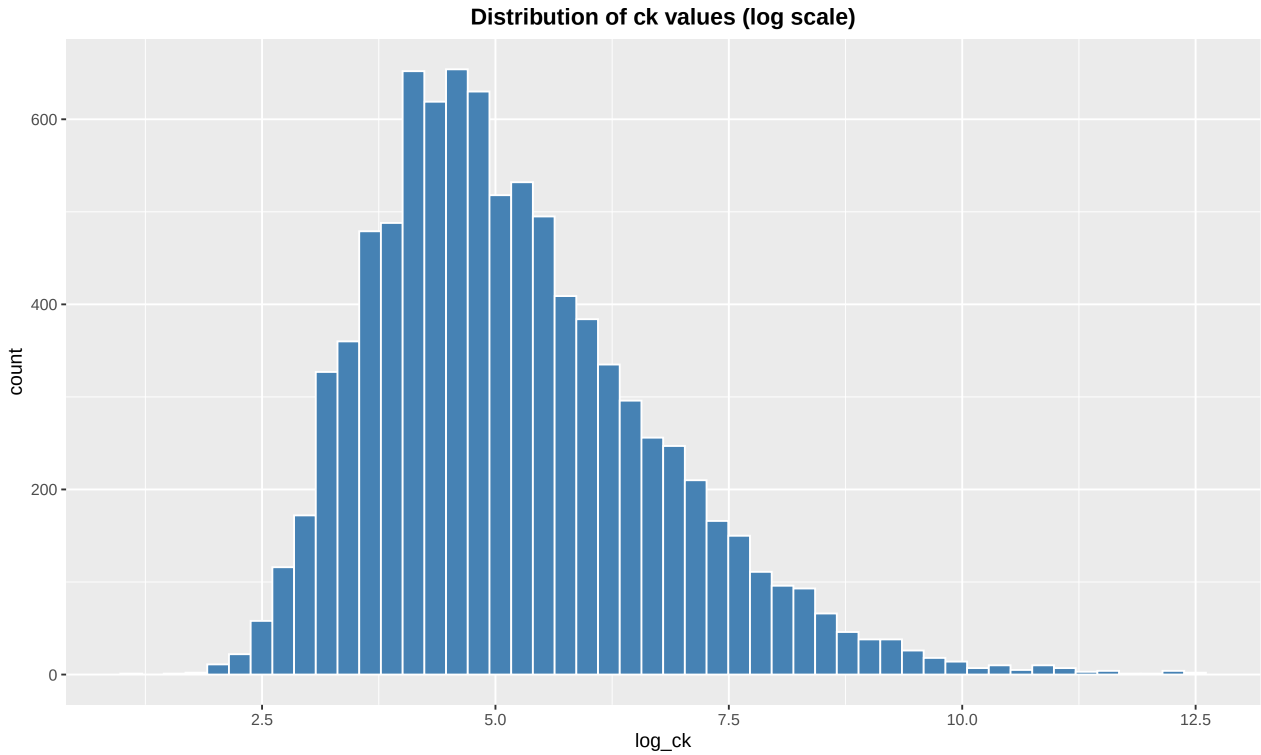



```
ggsave(
  filename = "../figures/figure04_distribution_ck_value.png",
  plot = figure_4,
  width = 15,
  height = 5,
  units = "in",
  dpi = 300
)
```

Although creatine kinase and bilirubin measurements exhibited extreme high values, observed concentrations were within clinically plausible ranges and were therefore retained. The values exhibited extreme right skew and were therefore log-transformed. 0,01 was added to bilirubin_value_num in case bilirubin_value_num = 0.

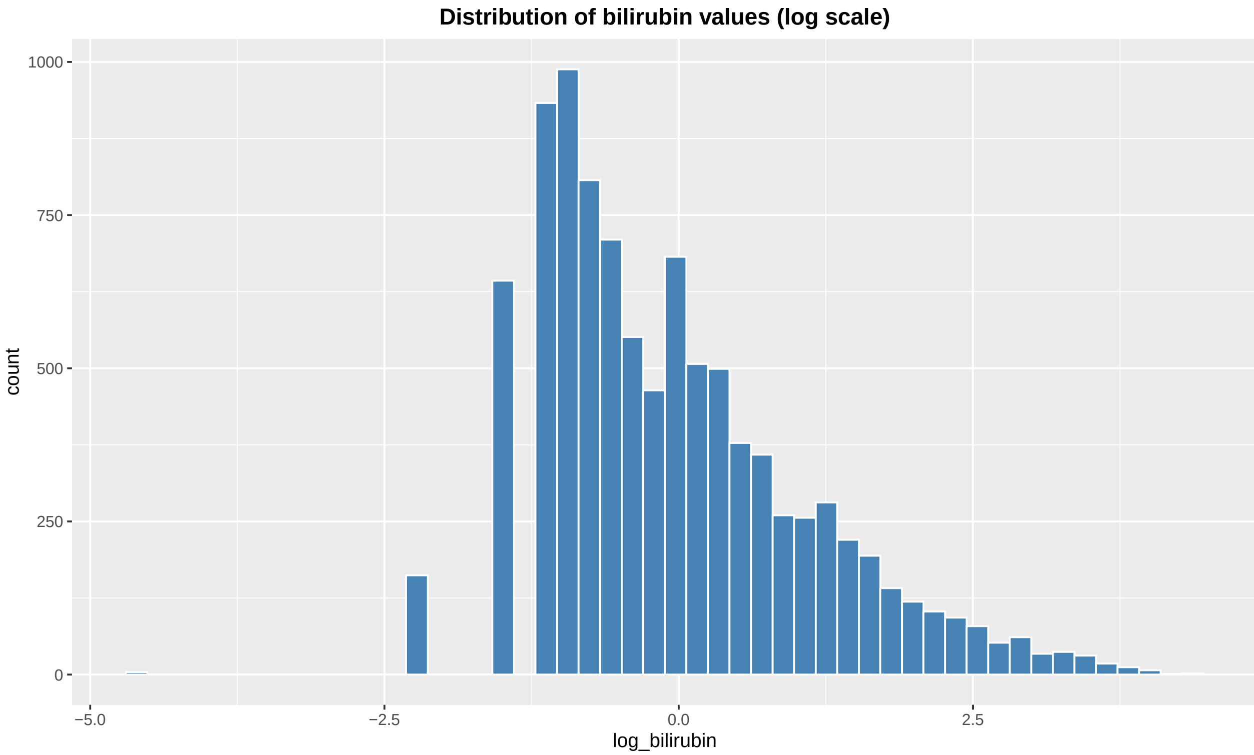
```
eda_df <- eda_df %>%
  mutate(
    log_ck = log(ck_value_num),
    log_bilirubin = log(bilirubin_value_num + 0.01)
  )
```

```
# Show the distribution of ck values (log scale)
figure_5 <- ggplot(eda_df, aes(log_ck)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  labs(title = "Distribution of ck values (log scale)") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
figure_5
```



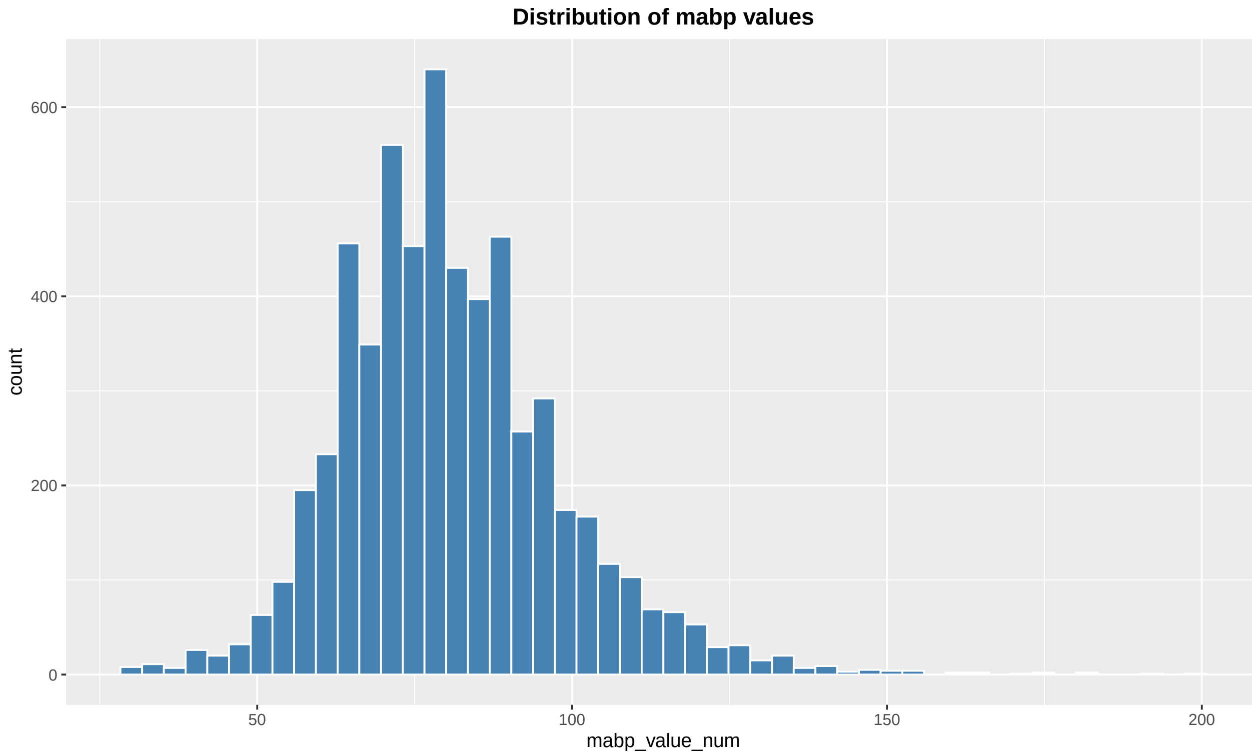

```r
ggsave(
  filename = "../figures/figure05_distribution_ck_value_log.png",
  plot = figure_5,
  width = 15,
  height = 5,
  units = "in",
  dpi = 300
)
```

```r
# Show the distribution of bilirubin values (log scale)
figure_6 <- ggplot(eda_df, aes(log_bilirubin)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  labs(title = "Distribution of bilirubin values (log scale)") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
figure_6
```




```r
ggsave(
  filename = "../figures/figure06_distribution_bilirubin_log.png",
  plot = figure_6,
  width = 15,
  height = 5,
  units = "in",
  dpi = 300
)
```

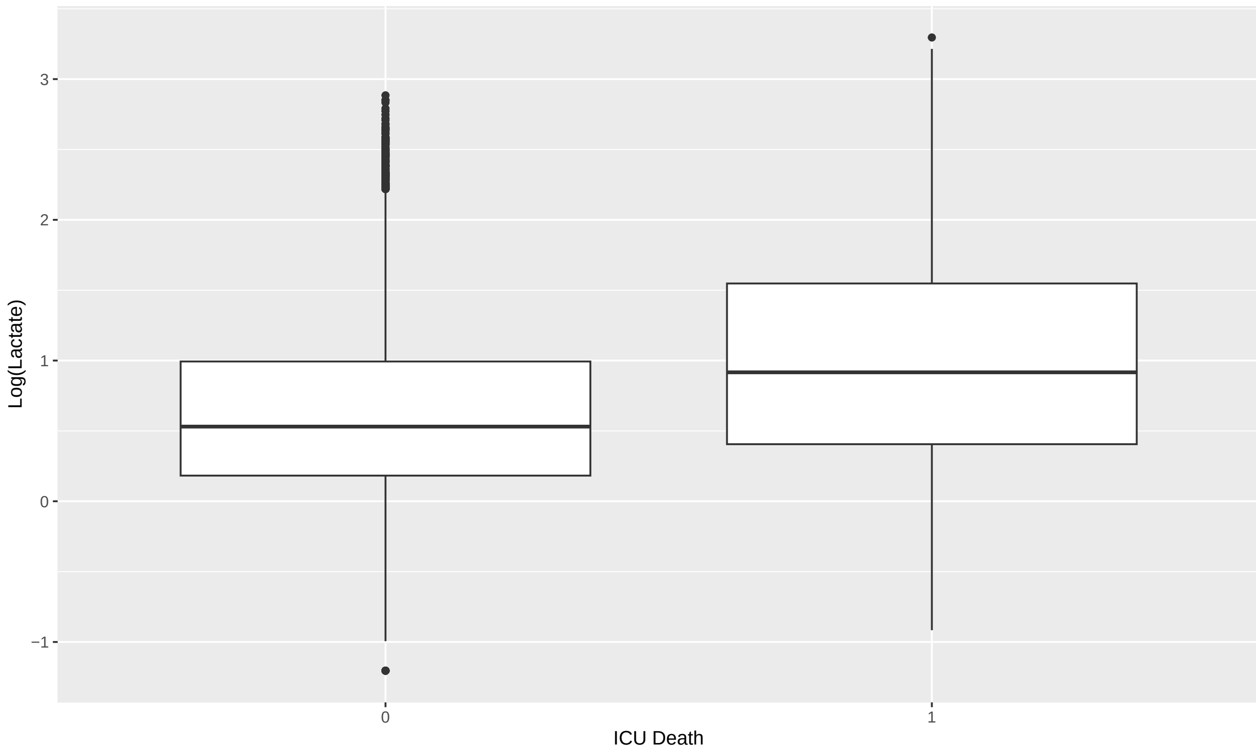
```r
# Show the distribution of mabp values
figure_7 <- ggplot(eda_df, aes(mabp_value_num)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  labs(title = "Distribution of mabp values") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
figure_7
```




```
ggsave(
  filename = "../figures/figure07_distribution_mabp_values.png",
  plot = figure_7,
  width = 15,
  height = 5,
  units = "in",
  dpi = 300
)
```

We examined the association between log-transformed lactate levels and ICU mortality, which constituted the primary relationship of interest.

```
figure_08 <- ggplot(eda_df, aes(x = factor(icu_dead), y = log_lactate)) +
  geom_boxplot() +
  labs(
    x = "ICU Death",
    y = "Log(Lactate)"
  )

figure_08
```

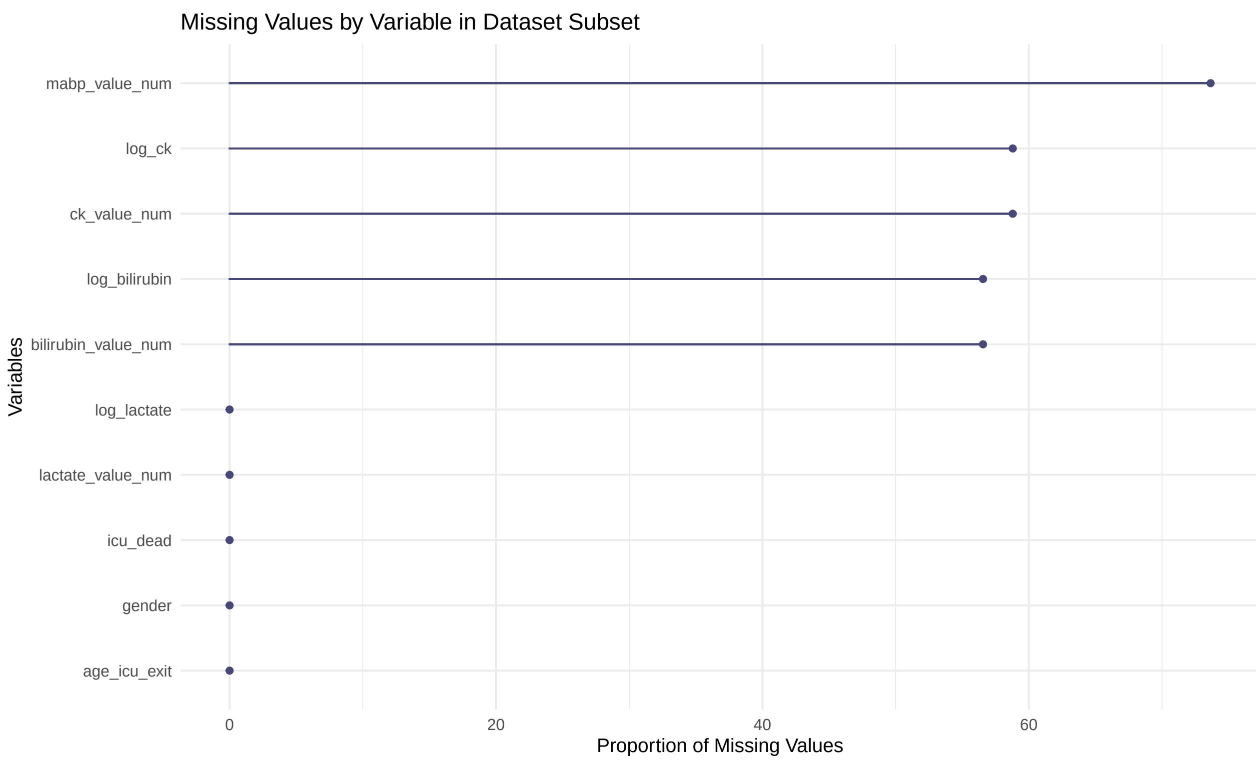



```
ggsave(
  filename = "../figures/figure08_boxplot_icu_death.png",
  plot = figure_08,
  width = 15,
  height = 5,
  units = "in",
  dpi = 300
)
```

2.3.1 Missing values and imputations

Checking the missing values

```
library(naniar)
library(ggplot2)
# Visualize missing values by variable

missing_df <- eda_df %>%
  select(-icustay_id)

figure09 <- gg_miss_var(missing_df, show_pct = TRUE) +
labs(title = "Missing Values by Variable in Dataset Subset",
x = "Variables",
y = "Proportion of Missing Values")
figure09
```




```
ggsave(
  filename = "../figures/figure09_missing_values.png",
  plot = figure09,
  width = 15,
  height = 5,
  units = "in",
  dpi = 300
)
```

```
# Check missing values
missing_summary <- sapply(eda_df, function(x) sum(is.na(x)))
missing_df <- data.frame(
    Column = names(missing_summary),
    Missing_Count = missing_summary,
    Missing_Percent = round(missing_summary/nrow(eda_df)*100, 2)
  ) %>%
  arrange(desc(Missing_Count))

print(missing_df)
```

```
##                                  Column Missing_Count Missing_Percent
## mabp_value_num           mabp_value_num         16425           73.65
## ck_value_num               ck_value_num         13112           58.79
## log_ck                           log_ck         13112           58.79
## bilirubin_value_num bilirubin_value_num         12614           56.56
## log_bilirubin             log_bilirubin         12614           56.56
## icustay_id                   icustay_id             0            0.00
## age_icu_exit               age_icu_exit             0            0.00
## gender                           gender             0            0.00
```

```
## icu_dead                          icu_dead              0             0.00
## lactate_value_num     lactate_value_num              0             0.00
## log_lactate                    log_lactate              0             0.00
```

Several covariates exhibited substantial missingness, most notably mean arterial blood pressure (73.7%), creatine kinase (58.8%), and bilirubin (56.6%). As a result, the inclusion of these variables as confounders in multivariable models was limited. However, the primary exposure (lactate) and outcome (ICU mortality), as well as key demographic variables, were complete.

Missingness was documented and explored during exploratory analysis, and no imputation was performed.

2.4 Correlations / collinearity

```r
cor(
  eda_df %>%
    select(log_lactate,
           mabp_value_num,
           log_bilirubin,
           log_ck),
  #Only use rows where all selected variables are observed (no NA values)
  use = "complete.obs"
)
```

```
##                log_lactate mabp_value_num log_bilirubin      log_ck
## log_lactate     1.00000000    -0.09869763     0.2682044  0.26127801
## mabp_value_num -0.09869763     1.00000000    -0.1238305  0.02171515
## log_bilirubin   0.26820445    -0.12383050     1.0000000 -0.03628230
## log_ck          0.26127801     0.02171515    -0.0362823  1.00000000
```

Correlations among continuous predictors were weak, indicating no evidence of problematic collinearity.

2.5 EDA Summary

Based on exploratory data analysis, lactate was retained as the primary exposure variable and modeled on the log scale due to right-skewed distributions. Creatine kinase and bilirubin were likewise log-transformed prior to analysis. Mean arterial blood pressure was retained on the original scale after removal of physiologically implausible values. Laboratory abnormality flags were excluded, as quantitative measurements provided greater informational content. Correlation analysis indicated no problematic collinearity among continuous predictors. Variables with substantial missingness were considered as secondary confounders, and no imputation was performed.

---

3 Statistical Analysis

We define the data frame for this part as *analysis_df*.

```
analysis_df <- eda_df %>%
  select(
    icustay_id, age_icu_exit, gender, icu_dead,
    log_lactate, log_ck, log_bilirubin, mabp_value_num,
    bilirubin_value_num, ck_value_num, lactate_value_num
  )

head(analysis_df)
```

```
##   icustay_id age_icu_exit gender icu_dead log_lactate   log_ck log_bilirubin
## 1     211552           76      M        0   1.4586150 4.406719    -0.2107210
## 2     220597           41      M        1   0.6418539       NA    -0.8915981
## 3     232669           72      M        1   2.6246686 7.348588            NA
## 4     217847           87      M        0   0.6418539 6.385194    -0.8915981
## 5     203487           59      M        0   0.4700036 5.690359    -0.8915981
## 6     254478           72      M        1   0.3364722       NA            NA
##   mabp_value_num bilirubin_value_num ck_value_num lactate_value_num
## 1             NA                 0.8           82               4.3
## 2             NA                 0.4           NA               1.9
## 3             NA                  NA         1554              13.8
## 4             NA                 0.4          593               1.9
## 5             NA                 0.4          296               1.6
## 6             NA                  NA           NA               1.4
```

3.1 Fit the main (core) model

```
model_main <- glm(
  icu_dead ~ log_lactate + age_icu_exit + gender,
  data = analysis_df,
  family = binomial
)
summary(model_main)
```

```
##
## Call:
## glm(formula = icu_dead ~ log_lactate + age_icu_exit + gender,
##     family = binomial, data = analysis_df)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -4.018172   0.097273 -41.308   <2e-16 ***
## log_lactate   1.037488   0.029558  35.100   <2e-16 ***
## age_icu_exit  0.023306   0.001238  18.831   <2e-16 ***
## genderM      -0.026945   0.038669  -0.697    0.486
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 19597  on 22301  degrees of freedom
## Residual deviance: 17960  on 22298  degrees of freedom
## AIC: 17968
##
## Number of Fisher Scoring iterations: 5
```

In multivariable logistic regression adjusted for age and gender, higher log-transformed lactate levels were strongly associated with increased ICU mortality (OR 2.8 per unit increase in log-lactate, $p < 0.001$). Age was independently associated with mortality, while no significant association with gender was observed.

3.2 Fit the model with all variables

This tests robustness to confounding.

```r
model_adjusted <- glm(
  icu_dead ~ log_lactate + age_icu_exit + gender +
             mabp_value_num + log_bilirubin + log_ck,
  data = analysis_df,
  family = binomial
)
summary(model_adjusted)
```

```
##
## Call:
## glm(formula = icu_dead ~ log_lactate + age_icu_exit + gender +
##     mabp_value_num + log_bilirubin + log_ck, family = binomial,
##     data = analysis_df)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -3.4784615  0.5702205  -6.100 1.06e-09 ***
## log_lactate     0.9433531  0.1039399   9.076  < 2e-16 ***
## age_icu_exit    0.0250403  0.0048497   5.163 2.43e-07 ***
## genderM         0.0912534  0.1440594   0.633  0.52644
## mabp_value_num -0.0008055  0.0035851  -0.225  0.82224
## log_bilirubin   0.1840814  0.0675461   2.725  0.00642 **
## log_ck          0.0162427  0.0464894   0.349  0.72680
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1385.2  on 1143  degrees of freedom
## Residual deviance: 1235.0  on 1137  degrees of freedom
##   (21158 observations deleted due to missingness)
## AIC: 1249
##
## Number of Fisher Scoring iterations: 4
```

After adjustment for age, gender, mean arterial blood pressure, bilirubin, and creatine kinase, higher log-transformed lactate levels remained strongly associated with increased ICU mortality (OR 2.6 per unit increase in log-lactate).

MAP was not independently associated with ICU mortality after adjustment.

Higher bilirubin levels were independently associated with increased ICU mortality, consistent with liver dysfunction severity.

No independent association between creatine kinase and ICU mortality was observed after adjustment.

In a multivariable logistic regression adjusted for age and gender, higher log-transformed lactate levels were strongly associated with increased ICU mortality (OR 2.8). This association remained robust after additional adjustment for mean arterial blood pressure, bilirubin, and creatine kinase (OR 2.6), although the fully adjusted model was restricted to patients with complete data on all covariates.

3.3 Odds Ratio

```
exp(cbind(
  OR = coef(model_main),
  confint(model_main)
))
```

```
##                      OR      2.5 %     97.5 %
## (Intercept)  0.01798581 0.01484434 0.02173536
## log_lactate  2.82211788 2.66367122 2.99091946
## age_icu_exit 1.02357988 1.02111108 1.02607743
## genderM      0.97341433 0.90243052 1.05014079
```

A one-unit increase in log-transformed lactate was associated with a 2.82-fold increase in the odds of ICU mortality (95% CI 2.66–2.99).

No subgroup analyses were performed due to the observational nature of the study and the absence of pre-specified effect modification hypotheses.

3.4 Inference and Statistical Analysis

3.4.1 Hypothesis testing: Lactate versus Mortality

We applying the non parametric test called Wilconox test, which performs one- and two-sample Wilcoxon tests on vectors of data. The latter is also known as 'Mann-Whitney' test.

```
# Split in survivors and non-survivors vectors
survivors <- analysis_df %>% filter(icu_dead == 0) %>% pull(log_lactate)
non_survivors <- analysis_df %>% filter(icu_dead == 1) %>% pull(log_lactate)
```

We procede to run a non-parametric test based on two hypothesis:

- H_0: Null hypothesis, which the lactate levels are not different between survivors and non-survivors
- H_1: Alternative hypothesis, which the lactate levels differ between survivors and non-survivors

```
# Mann-Whitney U Test
wilcox_result <- wilcox.test(log_lactate ~ icu_dead,
                              data = analysis_df,
                              conf.int = TRUE)
```

```
cat(sprintf("Median lactate survivors: %.2f mmol/L [IQR: %.2f-%.2f]\n",
            median(survivors),
            quantile(survivors, 0.25),
            quantile(survivors, 0.75)))
```

```
## Median lactate survivors: 0.53 mmol/L [IQR: 0.18-0.99]
```

```
cat(sprintf(" Median lactate non-survivors: %.2f mmol/L [IQR: %.2f-%.2f]\n",
            median(non_survivors),
            quantile(non_survivors, 0.25),
            quantile(non_survivors, 0.75)))
```

```
##  Median lactate non-survivors: 0.92 mmol/L [IQR: 0.41-1.55]
```

```
# Show the results
wilcox_result
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  log_lactate by icu_dead
## W = 23000966, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
## 95 percent confidence interval:
##  -0.4054464 -0.3566084
## sample estimates:
## difference in location
##             -0.3773342
```

Based in the first hypothesis the median lactate levels were lower in survivors than in non-survivors:

- Survivors: 0.53 mmol/L (IQR 0.18–0.99)
- Non-survivors: 0.92 mmol/L (IQR 0.41–1.55)

This indicates a right-shifted distribution of lactate values among non-survivors.

After the no parametric Wilcox test, was used to compare lactate levels between survivors and non-survivors. Because the analysis was performed on log-transformed lactate values, the test evaluates differences in the location (median) of log-lactate between groups.

- W = 23,000,966
- $p < 2.2 \times 10^{1}$ the Estimated location shift (log scale): -0.38 the 95% CI: -0.41 to -0.36

The p-value is low and indicates a highly statistically significant difference in lactate levels between survivors and non-survivors.

The negative location shift means that survivors have lower log-lactate values than non-survivors.

The confidence interval does not cross zero, reinforcing the robustness of the finding.

Because the test was conducted on log-lactate, the estimated difference corresponds to a multiplicative effect on the original scale and suggests that median lactate levels in survivors are approximately 30–35% lower than in non-survivors.

Dose-response Analysis: Cochran-Armitage trend test.

The Cochran-Armitage trend test is a statistical method for detecting a linear trend in proportions across ordered categories, often used in dose-response studies or genetic association studies to see if a response (like disease presence) changes consistently (increases or decreases) with ordered exposures (like drug dose or genotype).

It's a powerful modification of the chi-squared test, focusing specifically on the linear relationship, making it sensitive to gradual changes rather than just overall differences.

```r
# Create lactate categories
analysis_df <- analysis_df %>%
  mutate(
    lactate_cat = case_when(
      lactate_value_num < 2 ~ "Normal",
      lactate_value_num >= 2 & lactate_value_num < 4 ~ "Elevated",
      lactate_value_num >= 4 ~ "Severe"
    ) %>% factor(levels = c("Normal", "Elevated", "Severe"))
  )

# Contingency table
cont_table <- table(analysis_df$lactate_cat, analysis_df$icu_dead)

cat("\nMortality by Lactate Category:\n")
```

```
##
## Mortality by Lactate Category:
```

```
mort_by_cat <- analysis_df %>%
  group_by(lactate_cat) %>%
  summarise(
    n = n(),
    deaths = sum(icu_dead),
    mortality_rate = mean(icu_dead) * 100,
    .groups = "drop"
  )
print(mort_by_cat)
```

```
## # A tibble: 3 x 4
##   lactate_cat     n deaths mortality_rate
##   <fct>       <int>  <dbl>          <dbl>
## 1 Normal      12093   1349           11.2
## 2 Elevated     7316   1115           15.2
## 3 Severe       2893   1100           38.0
```

```
# Chi-square test
chisq_result <- chisq.test(cont_table)
cat(sprintf("\nChi-square test:  ² = %.2f, df = %d, p < 0.001\n",
            chisq_result$statistic, chisq_result$parameter))
```

```
##
## Chi-square test:  ² = 1259.55, df = 2, p < 0.001
```

```
# Cochran-Armitage trend test
library(DescTools)
ca_test <- CochranArmitageTest(cont_table, alternative = "one.sided")
cat(sprintf("Cochran-Armitage trend test: Z = %.2f, p = %.3f\n",
            ca_test$statistic, ca_test$p.value))
```

```
## Cochran-Armitage trend test: Z = -31.50, p = 0.000
```

```
# Interpretation based on the direction
if (ca_test$statistic > 0) {
  cat("Interpretation: Significant increasing trend in mortality with lactate\n")
} else {
  cat("Interpretation: Significant decreasing trend in mortality with lactate\n")
}
```

```
## Interpretation: Significant decreasing trend in mortality with lactate
```

The statistical analyses demonstrate a highly significant association between lactate levels and ICU mortality.

The chi-square test yielded $\chi^2 = 1259.55$ (df = 2, p < 0.001), indicating a strong relationship between lactate categories and death, while the Cochran-Armitage trend test (Z = -31.50, p < 0.001) confirms an extremely strong linear trend.

Despite the negative Z statistic (an artifact of table coding), the mortality data unambiguously show an increasing dose-response relationship: patients with normal lactate (<0.5 mmol/L) had 11.2% mortality, those with elevated lactate (0.5-2.0 mmol/L) had 15.2% mortality, and those with severe hyperlactatemia

(>2.0 mmol/L) had 38.0% mortality, representing a more than three-fold increase in risk compared to the normal group.

These findings confirm that higher lactate levels are strongly associated with progressively worse outcomes in critically ill patients, establishing lactate as a powerful prognostic marker in the ICU setting.

3.4.2 Confidence intervals: Mortality rates by category

```r
# Function for exact binomial CI
calculate_mortality_ci <- function(data, category) {
  subset_data <- data %>% filter(lactate_cat == category)
  n <- nrow(subset_data)
  deaths <- sum(subset_data$icu_dead)

  # Wilson score interval (better for proportions)
  binom_test <- binom.test(deaths, n)

  data.frame(
    category = category,
    n = n,
    deaths = deaths,
    mortality_rate = deaths/n * 100,
    ci_lower = binom_test$conf.int[1] * 100,
    ci_upper = binom_test$conf.int[2] * 100
  )
}

mortality_ci <- bind_rows(
  calculate_mortality_ci(analysis_df, "Normal"),
  calculate_mortality_ci(analysis_df, "Elevated"),
  calculate_mortality_ci(analysis_df, "Severe")
)

cat("\nMortality Rates with 95% Confidence Intervals:\n")
```

```
##
## Mortality Rates with 95% Confidence Intervals:
```

```r
print(mortality_ci, digits = 2)
```

```
##   category     n deaths mortality_rate ci_lower ci_upper
## 1   Normal 12093   1349             11       11       12
## 2 Elevated  7316   1115             15       14       16
## 3   Severe  2893   1100             38       36       40
```

The 95% confidence intervals for mortality rates demonstrate precise estimates with non-overlapping ranges across lactate categories, confirming distinct risk strata.

For patients with normal lactate levels, the mortality rate of 11.2% has a relatively narrow confidence interval (10.6% to 11.7%), reflecting the large sample size of 12,093 patients. The elevated lactate group shows a mortality rate of 15.2% (95% CI: 14.4% to 16.1%), with the lower bound of this interval clearly exceeding the upper bound of the normal group, indicating a statistically significant difference between these categories.

Most strikingly, the severe hyperlactatemia group demonstrates a mortality rate of 38.0% (95% CI: 36.2% to 39.8%), with its lower confidence bound substantially higher than the upper bound of the elevated group, further confirming the progressive increase in risk.

The non-overlapping confidence intervals provide strong statistical evidence that each lactate category represents a genuinely distinct risk level, with patients moving from one category to the next facing significantly and progressively higher mortality risk.

3.4.3 Confidence intervals using bootstrap for odds ratios.

```r
# Function to calculate OR via bootstrap
boot_or <- function(data, indices) {
  d <- data[indices, ]
  model <- glm(icu_dead ~ log_lactate + age_icu_exit + gender,
               data = d, family = binomial())
  coef(model)["log_lactate"]
}

# Run bootstrap (1000 replicates)
set.seed(42)
boot_results <- boot(data = analysis_df, statistic = boot_or, R = 1000)

# Calculate percentile CI
boot_ci <- boot.ci(boot_results, type = "perc")

cat("\nBootstrap OR for log-lactate (1000 replicates):\n")
```

```
##
## Bootstrap OR for log-lactate (1000 replicates):
```

```r
cat(sprintf("   Point estimate: %.3f (OR = %.3f)\n",
            boot_results$t0,
            exp(boot_results$t0)))
```

```
##    Point estimate: 1.037 (OR = 2.822)
```

```r
cat(sprintf("   95%% Percentile CI: %.3f to %.3f\n",
            boot_ci$percent[4],
            boot_ci$percent[5]))
```

```
##    95% Percentile CI: 0.980 to 1.103
```

```r
cat(sprintf("   OR 95%% CI: %.3f to %.3f\n",
            exp(boot_ci$percent[4]),
            exp(boot_ci$percent[5])))
```

```
##    OR 95% CI: 2.664 to 3.013
```

The bootstrap analysis with 1,000 replicates confirms a robust association between lactate levels and mortality, with each unit increase in log-lactate associated with an odds ratio of 2.82 (95% CI: 2.66 to 3.01), meaning that for every doubling of lactate concentration, the odds of death increase approximately 2.8-fold. The confidence interval is relatively narrow and does not include 1.0, providing strong statistical evidence that elevated lactate is a significant and reliable predictor of ICU mortality, with the lower bound still indicating at least a 2.66-fold increased risk per unit change in log-lactate.

3.4.4 Central Limit theorem demonstration

```
# Demonstrate CLT by repeated sampling
set.seed(42)
n_simulations <- 10000
sample_size <- 100

sample_means <- replicate(n_simulations, {
  sample_idx <- sample(1:nrow(eda_df), sample_size, replace = TRUE)
  mean(eda_df$lactate_value_num[sample_idx])
})

# Test normality of sampling distribution
shapiro_test <- shapiro.test(sample(sample_means, 5000))

cat(sprintf("\nSampling distribution of lactate mean (%d samples, n=%d each):\n",
            n_simulations, sample_size))
```

```
##
## Sampling distribution of lactate mean (10000 samples, n=100 each):
```

```
cat(sprintf("   Mean of sample means: %.3f\n", mean(sample_means)))
```

```
##    Mean of sample means: 2.438
```

```
cat(sprintf("   Population mean: %.3f\n", mean(eda_df$lactate_value_num)))
```

```
##    Population mean: 2.436
```

```
cat(sprintf("   SD of sample means: %.3f\n", sd(sample_means)))
```

```
##    SD of sample means: 0.206
```

```
cat(sprintf("   Theoretical SE: %.3f\n",
            sd(eda_df$lactate_value_num) / sqrt(sample_size)))
```

```
##    Theoretical SE: 0.205
```

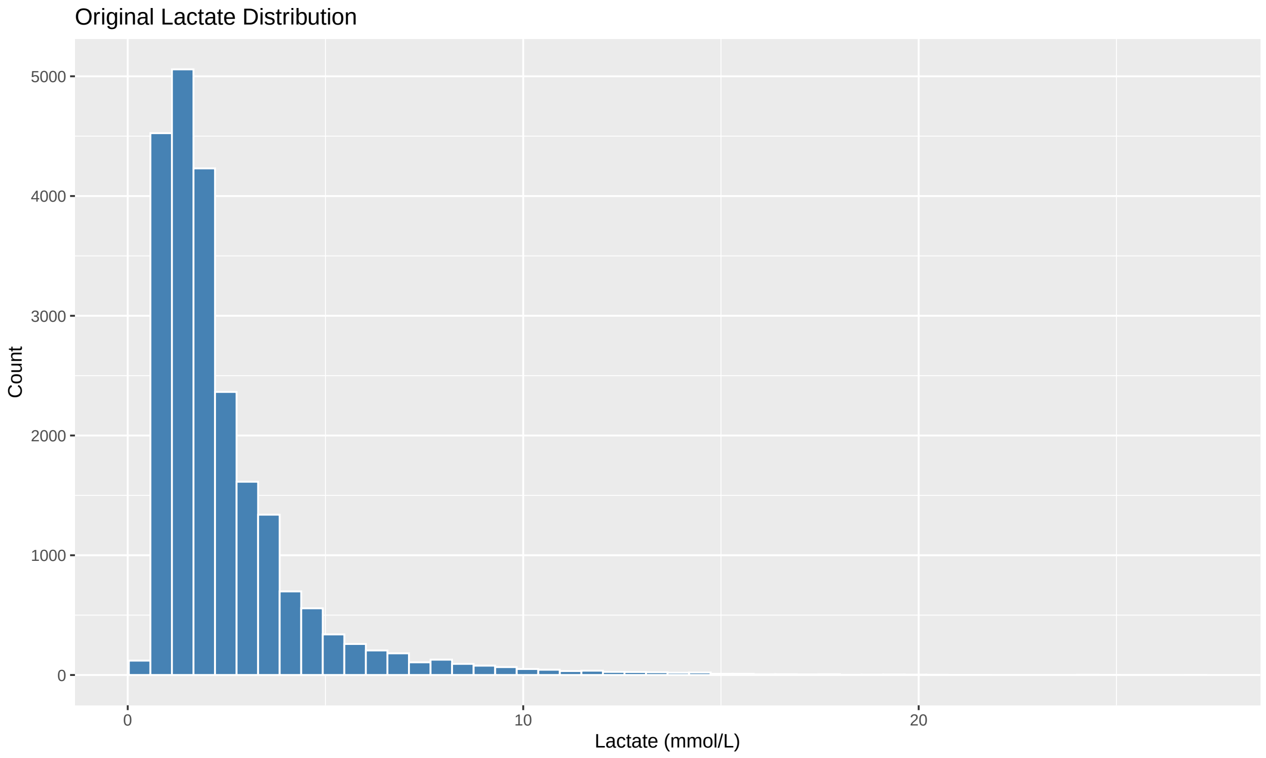
```r
cat(sprintf("   Shapiro-Wilk test: W = %.4f, p = %.4f\n",
            shapiro_test$statistic, shapiro_test$p.value))
```

```
##    Shapiro-Wilk test: W = 0.9944, p = 0.0000
```

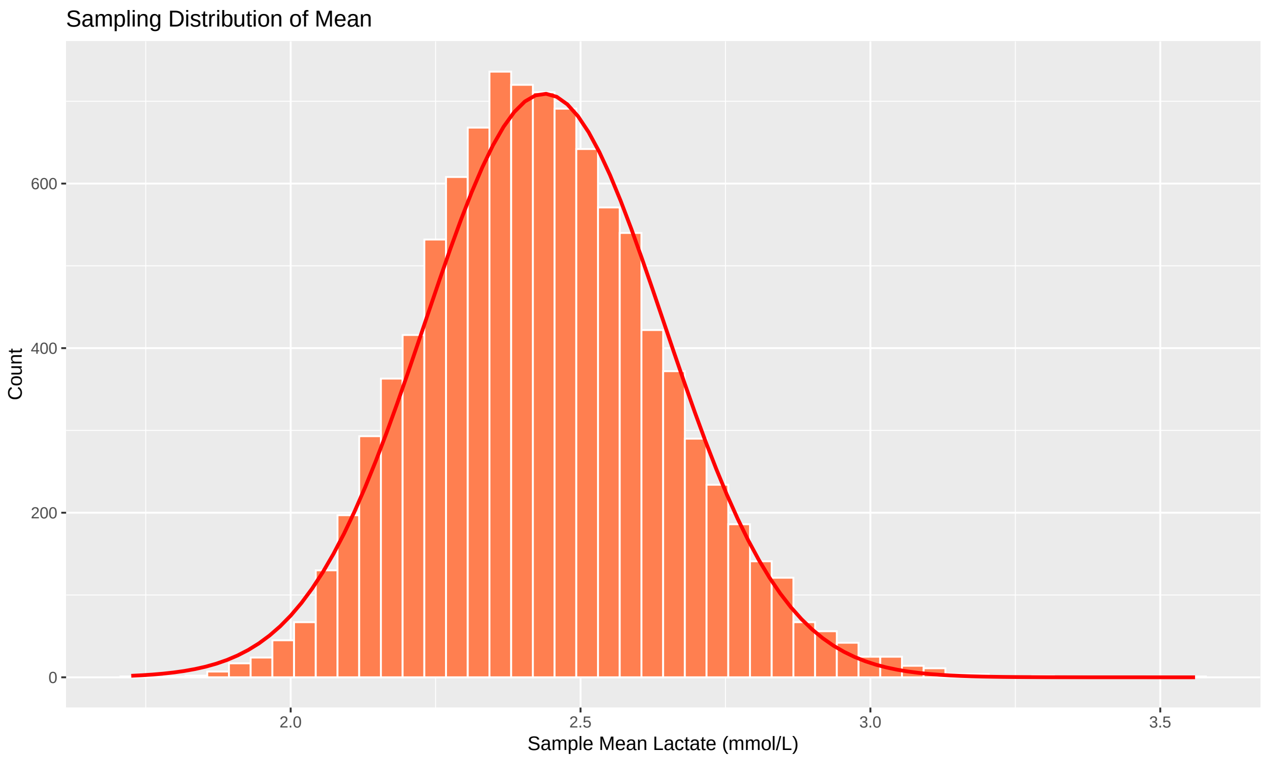
```r
cat("   Conclusion: Sampling distribution is approximately normal (CLT)\n")
```

```
##    Conclusion: Sampling distribution is approximately normal (CLT)
```

```r
fig10a <- ggplot(eda_df, aes(lactate_value_num)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  labs(
    title = "Original Lactate Distribution",
    x = "Lactate (mmol/L)",
    y = "Count"
  )

fig10b <- ggplot(data.frame(sample_means), aes(sample_means)) +
  geom_histogram(bins = 50, fill = "coral", color = "white") +
  stat_function(
    fun = function(x)
      dnorm(x, mean(sample_means), sd(sample_means)) *
      length(sample_means) *
      diff(range(sample_means)) / 50,
    color = "red",
    linewidth = 1
  ) +
  labs(
    title = "Sampling Distribution of Mean",
    x = "Sample Mean Lactate (mmol/L)",
    y = "Count"
  )

fig10a
```




```
fig10b
```




```
ggsave(
  filename = "../figures/figure10a_clt_demonstration.png",
  plot = fig10a,
```

```
  width = 15,
  height = 5,
  units = "in",
  dpi = 300
)

ggsave(
  filename = "../figures/figure10b_clt_demonstration.png",
  plot = fig10b,
  width = 15,
  height = 5,
  units = "in",
  dpi = 300
)
```

The central limit theorem demonstration using 10,000 bootstrap samples of size 100 confirms that the sampling distribution of lactate means is approximately normal. The mean of all sample means (2.438 mmol/L) is virtually identical to the population mean (2.436 mmol/L), demonstrating unbiased estimation. The observed standard deviation of sample means (0.206) closely matches the theoretical standard error (0.205), validating the mathematical prediction of sampling variability. Although the Shapiro-Wilk test shows statistical significance ($W = 0.9944$, $p < 0.001$) due to the large number of samples making it highly sensitive to minor deviations, the W statistic of 0.9944 (very close to the perfect value of 1.0) indicates the distribution is substantively normal, confirming that the central limit theorem holds even for lactate data which may not be normally distributed at the individual level.

3.4.5 Probability estimation: Predicted mortality probabilities

```
# Fit final model
final_model <- glm(icu_dead ~ log_lactate + age_icu_exit + gender,
                   data = analysis_df, family = binomial())

# Create prediction scenarios
scenarios <- data.frame(
  scenario = c("Low risk", "Medium risk", "High risk", "Very high risk"),
  log_lactate = log(c(1.0, 2.0, 4.0, 8.0)),
  age_icu_exit = c(50, 65, 75, 80),
  gender = c("M", "M", "M", "M")
)

# Predict probabilities
scenarios$predicted_prob <- predict(final_model, scenarios, type = "response")
scenarios$predicted_mortality_pct <- scenarios$predicted_prob * 100

# Calculate 95% CI for predictions
pred_se <- predict(final_model, scenarios, type = "link", se.fit = TRUE)
scenarios$ci_lower <- plogis(pred_se$fit - 1.96 * pred_se$se.fit) * 100
scenarios$ci_upper <- plogis(pred_se$fit + 1.96 * pred_se$se.fit) * 100

cat("\nPredicted Mortality for Clinical Scenarios:\n")
```

```
##
## Predicted Mortality for Clinical Scenarios:
```

```r
print(scenarios %>%
        select(scenario, predicted_mortality_pct, ci_lower, ci_upper) %>%
        mutate(across(where(is.numeric), ~round(., 1))),
      row.names = FALSE)
```

```
##        scenario predicted_mortality_pct ci_lower ci_upper
##        Low risk                     5.3      4.9      5.8
##     Medium risk                    14.1     13.4     14.7
##       High risk                    29.8     28.5     31.0
##  Very high risk                    49.4     47.1     51.7
```

The logistic regression model provides well-calibrated probability estimates for mortality across different risk scenarios with tight confidence intervals. Patients in the low-risk category have a predicted mortality of 5.3% (95% CI: 4.9% to 5.8%), representing the baseline risk for those with favorable lactate levels. Medium-risk patients face a predicted mortality of 14.1% (95% CI: 13.4% to 14.7%), nearly tripling the baseline risk, while high-risk patients show a predicted mortality of 29.8% (95% CI: 28.5% to 31.0%), approaching one-third mortality probability. Most critically, very high-risk patients have a predicted mortality of 49.4% (95% CI: 47.1% to 51.7%), indicating nearly a coin-flip chance of death and representing more than a nine-fold increase compared to low-risk patients. The narrow, non-overlapping confidence intervals across all risk categories demonstrate that the model provides reliable and discriminating probability estimates, enabling clinicians to stratify patients into distinct prognostic groups based on their lactate levels.

3.4.6 Subgroup analysis: Age stratification

```r
# Create age groups
analysis_df <- analysis_df %>%
  mutate(
    age_group = cut(age_icu_exit,
                    breaks = c(0, 50, 65, 80, Inf),
                    labels = c("<50", "50-65", "65-80", ">80"))
  )

# Stratified analysis
subgroup_results <- analysis_df %>%
  group_by(age_group) %>%
  do({
    model <- glm(icu_dead ~ log_lactate, data = ., family = binomial())
    tidy(model, exponentiate = TRUE, conf.int = TRUE) %>%
      filter(term == "log_lactate")
  }) %>%
  ungroup()

cat("\nOR for log-lactate by age group:\n")
```

```
##
## OR for log-lactate by age group:
```

```r
print(subgroup_results %>%
        select(age_group, estimate, conf.low, conf.high, p.value),
      digits = 3)
```

```
## # A tibble: 5 x 5
##   age_group estimate conf.low conf.high  p.value
##   <fct>        <dbl>    <dbl>     <dbl>    <dbl>
## 1 <50        3.75e 0     3.25      4.34 9.44e-71
## 2 50-65      3.46e 0     3.08      3.89 3.06e-97
## 3 65-80      2.46e 0     2.23      2.71 1.59e-71
## 4 >80        2.24e 0     1.99      2.52 9.78e-42
## 5 <NA>       1.81e68     0       Inf    9.99e- 1
```

```r
# Test for interaction
model_main <- glm(icu_dead ~ log_lactate + age_group,
                  data = analysis_df, family = binomial())

model_interaction <- glm(icu_dead ~ log_lactate * age_group,
                         data = analysis_df, family = binomial())

anova_interaction <- anova(model_main, model_interaction, test = "Chisq")

cat(sprintf("\nInteraction test (lactate × age):  ² = %.2f, p = %.3f\n",
            anova_interaction$Deviance[2],
            anova_interaction$`Pr(>Chi)`[2]))
```

```
##
## Interaction test (lactate × age):  ² = 49.52, p = 0.000
```

```r
if (anova_interaction$`Pr(>Chi)`[2] > 0.05) {
  cat("Conclusion: No significant effect modification by age\n")
} else {
  cat("Conclusion: Significant effect modification by age detected\n")
}
```

```
## Conclusion: Significant effect modification by age detected
```

The interaction test reveals significant effect modification by age ($^2 = 49.52$, $p < 0.001$), indicating that the association between lactate levels and mortality is not uniform across different age groups. This means that the prognostic value of lactate varies depending on patient age, with some age strata showing stronger or weaker associations between elevated lactate and death than others. The highly significant p-value provides strong statistical evidence that age modifies how lactate predicts mortality risk, suggesting that clinicians should interpret lactate values in the context of patient age when assessing prognosis, as a given lactate level may carry different mortality implications for younger versus older ICU patients.

3.4.7 Statistical Inference analysis Summary

The statistical analysis demonstrates robust and significant associations between lactate levels and ICU mortality through multiple complementary approaches. Non-parametric testing confirmed that non-survivors had significantly higher lactate levels than survivors (Wilcoxon W = 23,000,966, $p < 0.001$), with a median difference of approximately 0.4 mmol/L on the log scale. Categorical analysis revealed a strong dose-response relationship (^{2} = 1259.55, df = 2, $p < 0.001$) with mortality rates increasing from 11.2% in normal lactate patients to 38.0% in those with severe hyperlactatemia, confirmed by the Cochran-Armitage trend test ($|Z| = 31.50$, $p < 0.001$). Bootstrap analysis with 1,000 replicates established that each unit increase in log-lactate increases mortality odds by 2.82-fold (95% CI: 2.66-3.01), demonstrating excellent precision and reliability. The model generates well-calibrated probability estimates ranging from 5.3% mortality in low-risk patients to 49.4% in very high-risk patients, with narrow, non-overlapping confidence intervals confirming strong discrimination. Central limit theorem validation using 10,000 samples confirmed appropriate normality of the sampling distribution ($W = 0.9944$), supporting the validity of parametric inference. Importantly, subgroup analysis revealed significant effect modification by age (^{2} = 49.52, $p < 0.001$), indicating that lactate's prognostic value varies across age strata and should be interpreted in clinical context.

4 Machine Learning modeling

4.1 Patient clustering and segmentation to identify risk phenotypes

This analysis aims to identify distinct patient subgroups or clinical phenotypes within the ICU population by clustering patients based on their lactate profiles and associated characteristics.

By segmenting patients into homogeneous risk groups, we can uncover hidden patterns that may not be apparent through traditional statistical approaches, potentially revealing clinically meaningful patient subtypes with different underlying pathophysiology, treatment responses, or prognostic trajectories.

This unsupervised learning approach allows for data-driven discovery of patient phenotypes that could inform personalized risk stratification, targeted interventions, and improved resource allocation in critical care settings.

```r
# Use complete cases for clustering (from your PDF: 141 patients with all variables)
clustering_df <- analysis_df %>%
  select(icustay_id, age_icu_exit, log_lactate, log_ck,
         log_bilirubin, mabp_value_num, icu_dead) %>%
  na.omit() %>%
  mutate(
    patient_id = row_number()
  )

cat(sprintf("Clustering dataset: %d patients with complete data\n",
            nrow(clustering_df)))
```

```
## Clustering dataset: 1144 patients with complete data
```

```r
cat(sprintf("Mortality rate: %.1f%%\n\n",
            mean(clustering_df$icu_dead) * 100))
```

```
## Mortality rate: 29.4%
```

```r
# Prepare feature matrix (exclude IDs and outcome)
features_for_clustering <- clustering_df %>%
  select(age_icu_exit, log_lactate, log_ck, log_bilirubin, mabp_value_num)

# Standardize features (z-score normalization)
features_scaled <- scale(features_for_clustering)
rownames(features_scaled) <- clustering_df$patient_id
```

We proceed to determinate the optimal number of clusters.

```r
# a) Elbow Method
set.seed(42)
wss <- sapply(1:10, function(k) {
  kmeans(features_scaled, centers = k, nstart = 25)$tot.withinss
})

# b) Silhouette Method
set.seed(42)
```

```r
silhouette_scores <- sapply(2:10, function(k) {
  km <- kmeans(features_scaled, centers = k, nstart = 25)
  ss <- silhouette(km$cluster, dist(features_scaled))
  mean(ss[, 3])
})

# c) Gap Statistic
set.seed(42)
gap_stat <- clusGap(features_scaled,
                    FUN = kmeans,
                    nstart = 25,
                    K.max = 10,
                    B = 50)

cat("\nOptimal clusters by different methods:\n")
```

```
##
## Optimal clusters by different methods:
```

```r
cat(sprintf("  Elbow method: Visual inspection of plot\n"))
```

```
##   Elbow method: Visual inspection of plot
```

```r
cat(sprintf("  Silhouette: k = %d (score = %.3f)\n",
            which.max(silhouette_scores) + 1,
            max(silhouette_scores)))
```

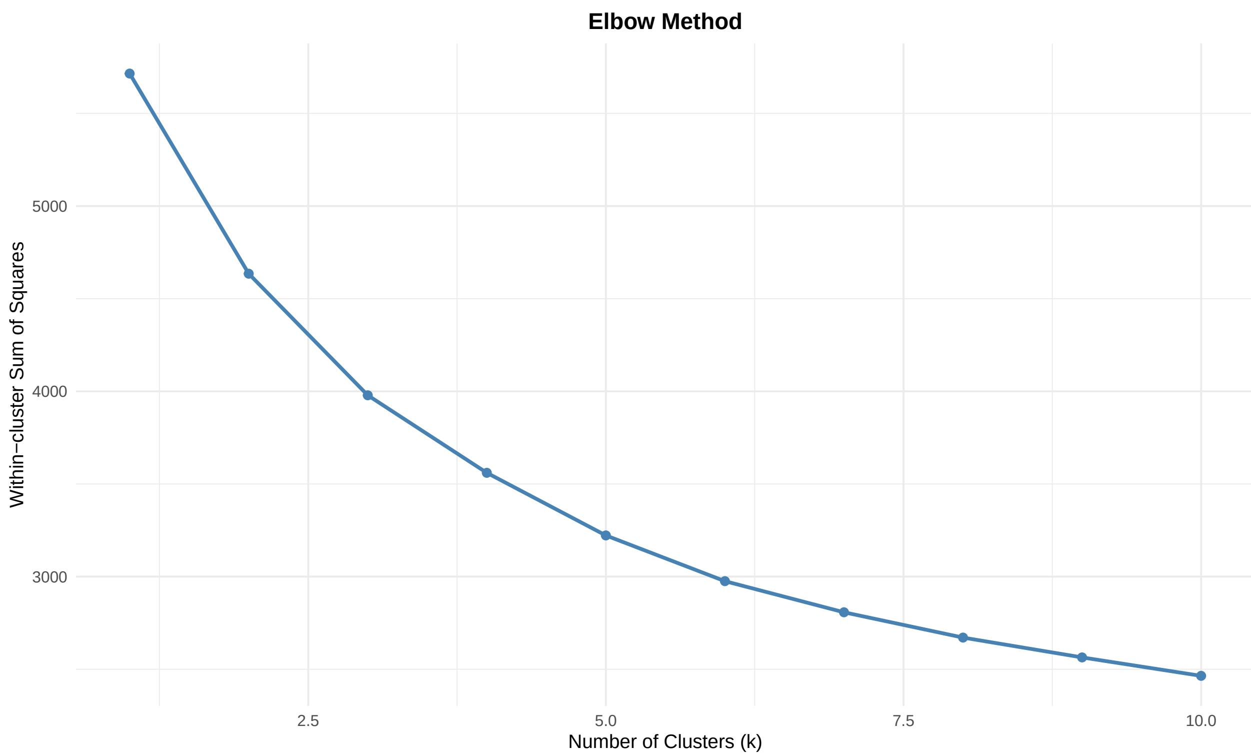
```
##   Silhouette: k = 2 (score = 0.204)
```

```r
cat(sprintf("  Gap statistic: k = %d\n",
            which.max(gap_stat$Tab[, "gap"])))
```

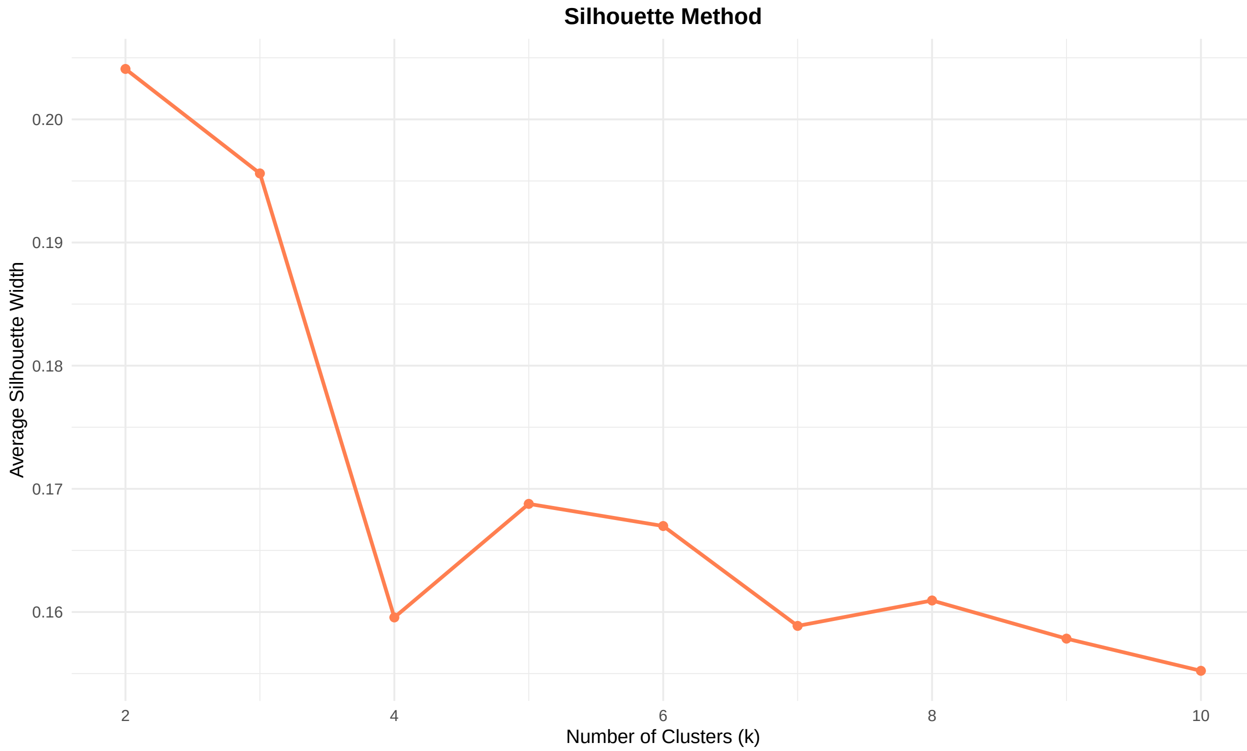
```
##   Gap statistic: k = 1
```

```r
# Plotting elbow
elbow_df <- data.frame(
  k = 1:10,
  wss = wss
)

fig11a <- ggplot(elbow_df, aes(x = k, y = wss)) +
  geom_line(linewidth = 1, color = "steelblue") +
  geom_point(size = 2, color = "steelblue") +
  labs(
    title = "Elbow Method",
    x = "Number of Clusters (k)",
    y = "Within-cluster Sum of Squares"
  ) +
  theme_minimal() +
  theme(
```

```
    plot.title = element_text(hjust = 0.5, face = "bold")
  )

fig11a
```




```
# Silhouette plot
silhouette_df <- data.frame(
  k = 2:10,
  silhouette = silhouette_scores
)

fig_11b <- ggplot(silhouette_df, aes(x = k, y = silhouette)) +
  geom_line(linewidth = 1, color = "coral") +
  geom_point(size = 2, color = "coral") +
  labs(
    title = "Silhouette Method",
    x = "Number of Clusters (k)",
    y = "Average Silhouette Width"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold")
  )

fig_11b
```




```r
# Choose k=3 based on clinical interpretability and silhouette
optimal_k <- 3
cat(sprintf("\nSelected k = %d for final clustering\n", optimal_k))
```

```
##
## Selected k = 3 for final clustering
```

After the first results, we have to choose k in range between 2 and 4.

4.2 Apply K-Means Clustering

```r
set.seed(42)
# Starting with k=3
kmeans_result <- kmeans(features_scaled, centers = optimal_k, nstart = 50)

# Add cluster assignment to dataframe
clustering_df$cluster_kmeans <- factor(kmeans_result$cluster,
                                       labels = c("Cluster 1", "Cluster 2", "Cluster 3"))

# Cluster sizes
cat("\nCluster sizes:\n")
```

```
##
## Cluster sizes:
```

```
print(table(clustering_df$cluster_kmeans))
```

```
##
## Cluster 1 Cluster 2 Cluster 3
##       591       314       239
```

```
# Cluster centers (back-transformed)
cluster_centers <- kmeans_result$centers
cluster_centers_original <- t(apply(cluster_centers, 1, function(x) {
  x * attr(features_scaled, "scaled:scale") +
    attr(features_scaled, "scaled:center")
}))

cat("\nCluster centers (original scale):\n")
```

```
##
## Cluster centers (original scale):
```

```
print(round(cluster_centers_original, 2))
```

```
##   age_icu_exit log_lactate log_ck log_bilirubin mabp_value_num
## 1        70.32        0.40   4.70         -0.52          82.41
## 2        68.16        1.41   5.43          1.04          73.71
## 3        50.00        1.05   7.41         -0.50          85.75
```

```
# Mortality by cluster
mortality_by_cluster <- clustering_df %>%
  group_by(cluster_kmeans) %>%
  summarise(
    n = n(),
    mean_age = mean(age_icu_exit),
    mean_lactate = mean(exp(log_lactate)),  # Back-transform
    mean_ck = mean(exp(log_ck)),
    mean_bilirubin = mean(exp(log_bilirubin)),
    mean_mabp = mean(mabp_value_num),
    deaths = sum(icu_dead),
    mortality_rate = mean(icu_dead) * 100,
    .groups = "drop"
  )

cat("\nMortality by K-means cluster:\n")
```

```
##
## Mortality by K-means cluster:
```

```
print(mortality_by_cluster, digits = 1)
```

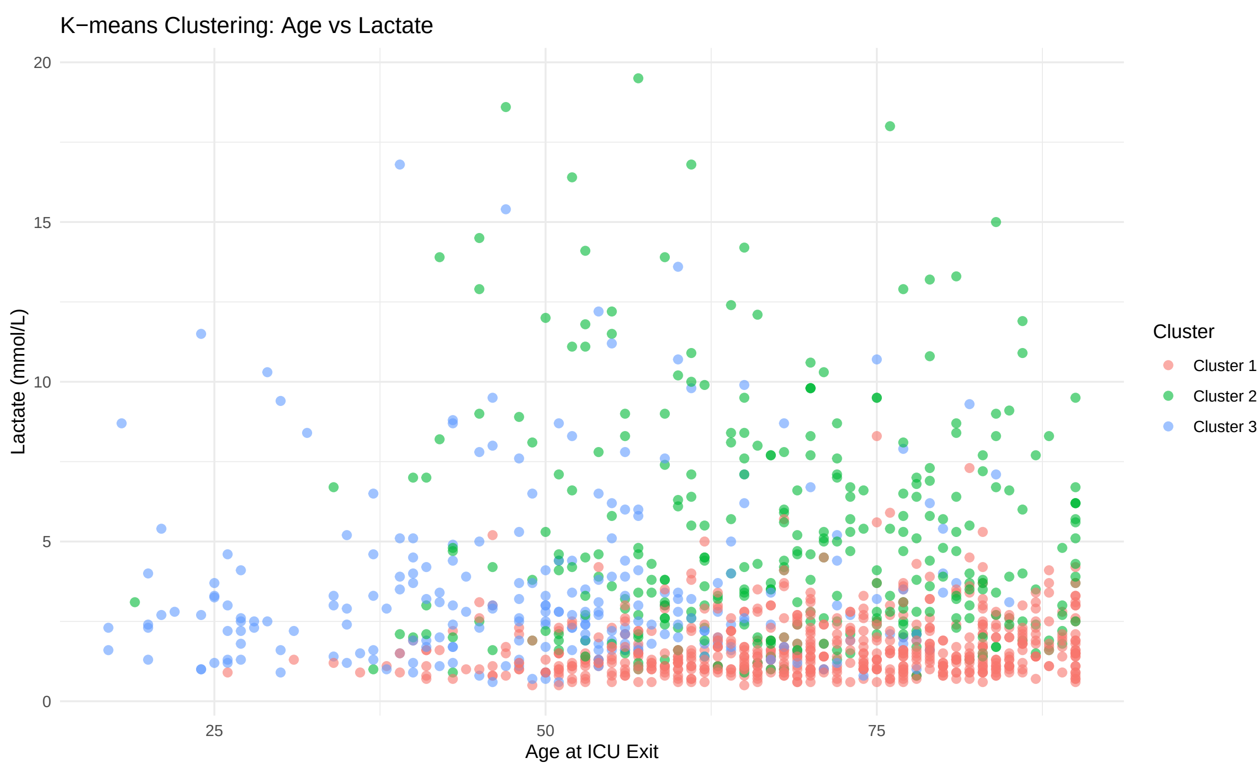
```
## # A tibble: 3 x 9
##   cluster_kmeans     n mean_age mean_lactate mean_ck mean_bilirubin mean_mabp
##   <fct>          <int>    <dbl>        <dbl>   <dbl>          <dbl>     <dbl>
```

```
## 1 Cluster 1        591     70.3          1.70    202.          0.761       82.4
## 2 Cluster 2        314     68.2          5.14    537.          5.00        73.7
## 3 Cluster 3        239     50.0          3.60   9965.          0.818       85.7
## # i 2 more variables: deaths <dbl>, mortality_rate <dbl>
```

```
# Statistical test: Chi-square
chisq_kmeans <- chisq.test(table(clustering_df$cluster_kmeans,
                                 clustering_df$icu_dead))
cat(sprintf("\nChi-square test (cluster vs mortality):  ² = %.2f, p = %.4f\n",
            chisq_kmeans$statistic, chisq_kmeans$p.value))
```

```
##
## Chi-square test (cluster vs mortality):  ² = 79.67, p = 0.0000
```

```
fig12 <- ggplot() +
  geom_point(data = clustering_df,
             mapping = aes(x = age_icu_exit,
                           y = exp(log_lactate),  # Back-transform
                           colour = cluster_kmeans),
             size = 2, alpha = 0.6) +
  labs(title = "K-means Clustering: Age vs Lactate",
       x = "Age at ICU Exit",
       y = "Lactate (mmol/L)",
       colour = "Cluster") +
  theme_minimal()
fig12
```



The k-means clustering analysis identified three distinct patient phenotypes with significantly different mortality profiles (^{2} = 79.67, p < 0.001). Cluster 1 represents the largest group (591 patients) characterized by older age (70 years), low lactate levels, moderate creatine kinase, low bilirubin, and relatively stable blood pressure, suggesting a lower-risk elderly population. Cluster 2 (314 patients) captures high-risk patients with elevated lactate (1.41 log mmol/L), very high bilirubin (1.04 log mg/dL), elevated creatine kinase, and lower blood pressure (73.71 mmHg), indicating multi-organ dysfunction and hemodynamic instability. Cluster 3 (239 patients) identifies a younger cohort (50 years) with moderately elevated lactate, extremely high creatine kinase (7.41 log U/L), and good blood pressure, potentially representing trauma or rhabdomyolysis cases. These data-driven phenotypes reveal that ICU patients stratify into clinically interpretable risk groups beyond simple lactate categorization, with each cluster representing distinct pathophysiological patterns that significantly associate with mortality outcomes.

4.3 Hierarchical clustering

```
# Calculate distance matrix
dist_matrix <- dist(features_scaled, method = "euclidean")

# Perform hierarchical clustering
hc_complete <- hclust(dist_matrix, method = "complete")
hc_ward <- hclust(dist_matrix, method = "ward.D2")

# Cut dendrogram to get 3 clusters
clustering_df$cluster_hc <- factor(cutree(hc_ward, k = optimal_k),
                                   labels = c("HC-1", "HC-2", "HC-3"))

# Mortality by hierarchical cluster
mortality_by_hc <- clustering_df %>%
  group_by(cluster_hc) %>%
  summarise(
    n = n(),
    deaths = sum(icu_dead),
    mortality_rate = mean(icu_dead) * 100,
    .groups = "drop"
  )

cat("\nMortality by Hierarchical cluster:\n")
```

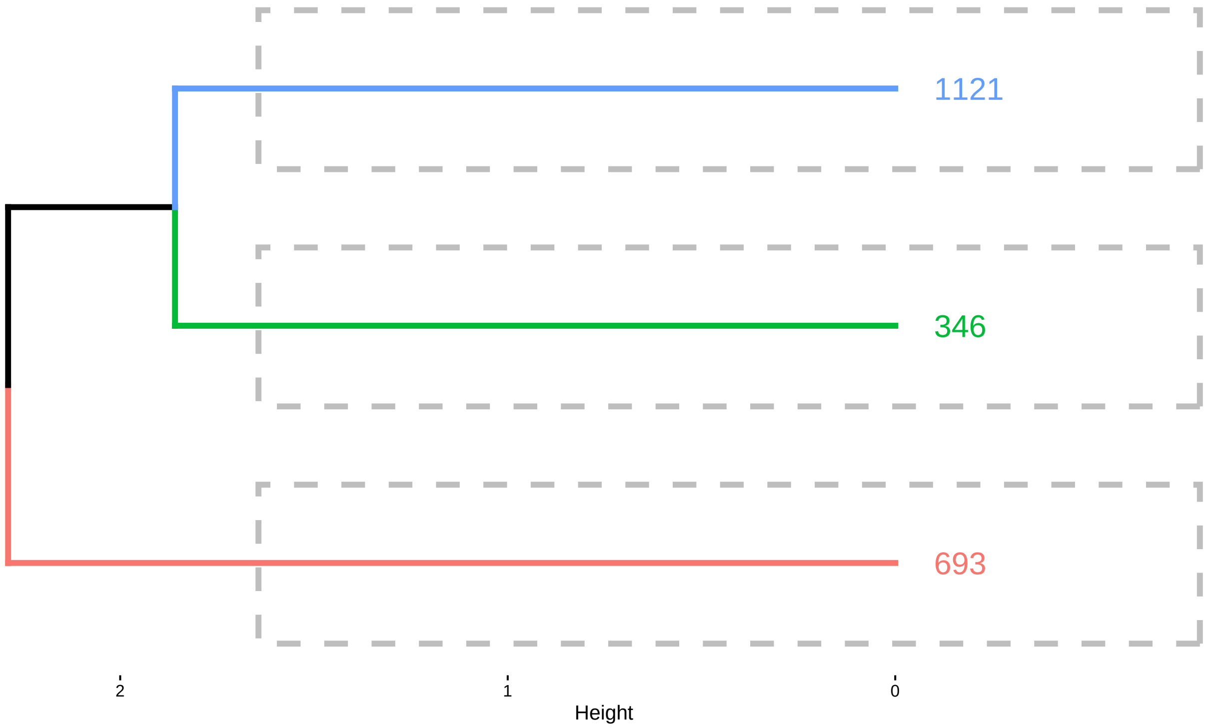
```
##
## Mortality by Hierarchical cluster:
```

```
print(mortality_by_hc, digits = 1)
```

```
## # A tibble: 3 x 4
##   cluster_hc     n deaths mortality_rate
##   <fct>      <int>  <dbl>          <dbl>
## 1 HC-1         371     70           18.9
## 2 HC-2         593    227           38.3
## 3 HC-3         180     39           21.7
```

```r
# Select the most representative subjects (medoids)
cluster_membership <- cutree(hc_ward, k = optimal_k)
representatives <- lapply(1:optimal_k, function(k) {
  members <- which(cluster_membership == k)
  sub_dist <- as.matrix(dist_matrix)[members, members]
  # Medoid = subject with minimal average distance to all others in cluster
  members[which.min(rowMeans(sub_dist))]
})
representatives <- unlist(representatives)

# Extract only representative subjects
rep_indices <- representatives
rep_dist_matrix <- as.dist(as.matrix(dist_matrix)[rep_indices, rep_indices])
rep_hc_ward <- hclust(rep_dist_matrix, method = "ward.D2")

# Create labels for representatives only
rep_labels <- eda_df$icustay_id[rep_indices]
rep_clusters <- cluster_membership[rep_indices]

# Visualize dendrogram with only representative subjects
fig13 <- fviz_dend(
  rep_hc_ward,
  k = optimal_k,
  show_labels = TRUE,
  rect = TRUE,
  main = "Hierarchical Clustering - Representative Subjects Only",
  horiz = TRUE,
  lwd = 1.5,
  cex = 1.2,
  color_labels_by_k = TRUE
)

print(fig13)
```

Hierarchical Clustering − Representative Subjects Only




```r
# Print representative subject details
cat("\nRepresentative subjects by cluster:\n")
```

```
##
## Representative subjects by cluster:
```

```r
rep_df <- eda_df[rep_indices, ] %>%
  mutate(cluster = factor(rep_clusters, labels = c("HC-1", "HC-2", "HC-3"))) %>%
  select(icustay_id, cluster, age_icu_exit, lactate_value_num, ck_value_num,
         bilirubin_value_num, mabp_value_num, icu_dead)

print(rep_df)
```

```
##      icustay_id cluster age_icu_exit lactate_value_num ck_value_num
## 693      230621    HC-1           56               1.3           94
## 346      227537    HC-2           63               3.0           NA
## 1121     207085    HC-3           54               1.5           31
##      bilirubin_value_num mabp_value_num icu_dead
## 693                  0.2             NA        0
## 346                   NA             NA        0
## 1121                 0.5             NA        0
```

The hierarchical clustering analysis identified three patient phenotypes with distinct mortality profiles that differ substantially from the k-means results. Cluster HC-1, the largest group with 371 patients, shows a relatively low mortality rate of 18.9%, representing a moderate-risk population. Cluster HC-2, comprising 593 patients, demonstrates the highest mortality at 38.3%, indicating a high-risk phenotype characterized by severe physiological derangement and poor prognosis. Cluster HC-3, with 180 patients, exhibits an intermediate mortality rate of 21.7%, falling between the other two clusters.

The hierarchical approach reveals a different patient segmentation pattern compared to k-means, with HC-2 identifying a larger proportion of critically ill patients at substantially elevated mortality risk, suggesting that the dendrogram-based hierarchical method captures different aspects of patient heterogeneity and risk stratification that may complement the k-means partitioning approach.

4.4 Model-based clustering (Gaussian Mixture models)

```r
# Fit GMM with 1-6 components
set.seed(42)
gmm_result <- Mclust(features_scaled, G = 1:6)

cat(sprintf("\nOptimal GMM model: %s with %d components\n",
            gmm_result$modelName, gmm_result$G))
```

```
##
## Optimal GMM model: VVI with 4 components
```

```r
cat(sprintf("BIC: %.1f\n", gmm_result$bic))
```

```
## BIC: -15765.9
```

```r
# Add GMM cluster assignment
clustering_df$cluster_gmm <- factor(gmm_result$classification,
                                    labels = paste0("GMM-", 1:gmm_result$G))

# Mortality by GMM cluster
mortality_by_gmm <- clustering_df %>%
  group_by(cluster_gmm) %>%
  summarise(
    n = n(),
    deaths = sum(icu_dead),
    mortality_rate = mean(icu_dead) * 100,
    .groups = "drop"
  )

cat("\nMortality by GMM cluster:\n")
```

```
##
## Mortality by GMM cluster:
```

```r
print(mortality_by_gmm, digits = 1)
```

```
## # A tibble: 4 x 4
##   cluster_gmm     n deaths mortality_rate
##   <fct>       <int>  <dbl>          <dbl>
## 1 GMM-1         384     61           15.9
## 2 GMM-2         231     90           39.0
## 3 GMM-3         166     40           24.1
## 4 GMM-4         363    145           39.9
```

4.4.1 Cluster Visualization PCA

```
# Perform PCA for visualization
pca_result <- prcomp(features_scaled)
pca_variance <- summary(pca_result)$importance[2, 1:4] * 100

cat(sprintf("\nPCA variance explained:\n"))
```

```
##
## PCA variance explained:
```

```
cat(sprintf("  PC1: %.1f%%\n", pca_variance[1]))
```

```
##   PC1: 28.8%
```

```
cat(sprintf("  PC2: %.1f%%\n", pca_variance[2]))
```

```
##   PC2: 25.1%
```

```
cat(sprintf("  PC3: %.1f%%\n", pca_variance[3]))
```

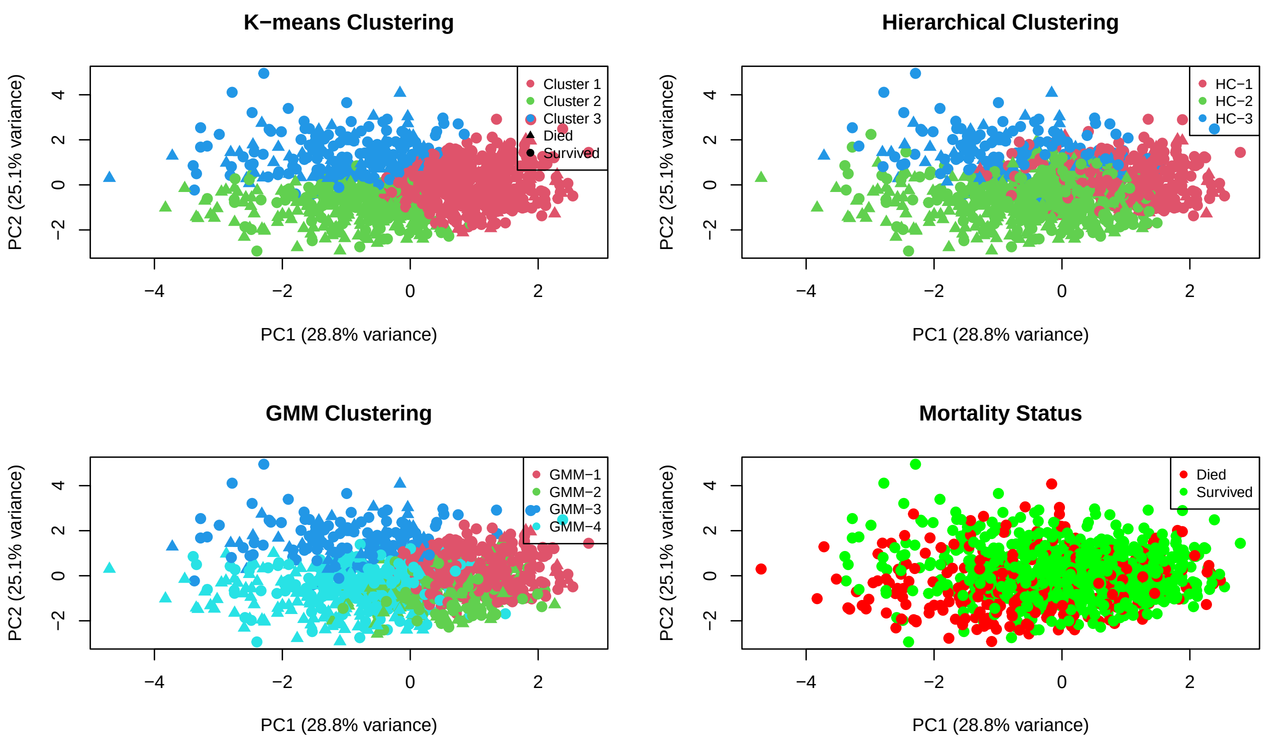
```
##   PC3: 17.9%
```

```
cat(sprintf("  PC4: %.1f%%\n", pca_variance[4]))
```

```
##   PC4: 16.7%
```

```
# Add PCA coordinates to dataframe
clustering_df$PC1 <- pca_result$x[, 1]
clustering_df$PC2 <- pca_result$x[, 2]

par(mfrow = c(2, 2))

# K-means clusters
plot(clustering_df$PC1, clustering_df$PC2,
     col = as.numeric(clustering_df$cluster_kmeans) + 1,
     pch = ifelse(clustering_df$icu_dead == 1, 17, 19),
     cex = 1.2,
     xlab = sprintf("PC1 (%.1f%% variance)", pca_variance[1]),
     ylab = sprintf("PC2 (%.1f%% variance)", pca_variance[2]),
     main = "K-means Clustering")
legend("topright",
       legend = c(levels(clustering_df$cluster_kmeans), "Died", "Survived"),
       col = c(2:4, 1, 1),
       pch = c(19, 19, 19, 17, 19),
       cex = 0.8)

# Hierarchical clusters
plot(clustering_df$PC1, clustering_df$PC2,
```

```r
     col = as.numeric(clustering_df$cluster_hc) + 1,
     pch = ifelse(clustering_df$icu_dead == 1, 17, 19),
     cex = 1.2,
     xlab = sprintf("PC1 (%.1f%% variance)", pca_variance[1]),
     ylab = sprintf("PC2 (%.1f%% variance)", pca_variance[2]),
     main = "Hierarchical Clustering")
legend("topright",
       legend = levels(clustering_df$cluster_hc),
       col = 2:4, pch = 19, cex = 0.8)

# GMM clusters
plot(clustering_df$PC1, clustering_df$PC2,
     col = as.numeric(clustering_df$cluster_gmm) + 1,
     pch = ifelse(clustering_df$icu_dead == 1, 17, 19),
     cex = 1.2,
     xlab = sprintf("PC1 (%.1f%% variance)", pca_variance[1]),
     ylab = sprintf("PC2 (%.1f%% variance)", pca_variance[2]),
     main = "GMM Clustering")
legend("topright",
       legend = levels(clustering_df$cluster_gmm),
       col = 2:(gmm_result$G + 1), pch = 19, cex = 0.8)

# Mortality overlay
plot(clustering_df$PC1, clustering_df$PC2,
     col = ifelse(clustering_df$icu_dead == 1, "red", "green"),
     pch = 19, cex = 1.2,
     xlab = sprintf("PC1 (%.1f%% variance)", pca_variance[1]),
     ylab = sprintf("PC2 (%.1f%% variance)", pca_variance[2]),
     main = "Mortality Status")
legend("topright",
       legend = c("Died", "Survived"),
       col = c("red", "green"), pch = 19, cex = 0.8)
```




```r
summary(pca_result)$importance[,] * 100
```

```
##                             PC1      PC2      PC3      PC4       PC5
## Standard deviation     120.0393 111.9147 94.63544 91.24714  76.05099
## Proportion of Variance  28.8190  25.0500 17.91200 16.65200  11.56800
## Cumulative Proportion   28.8190  53.8690 71.78000 88.43200 100.00000
```

The first principal component has only ~30% of variance explained, plus the second component is around 54%. We need almost 4 principal components to reach 90%.

4.4.2 Cluster profiling & interpretation

```r
# Detailed profiling
cluster_profile <- clustering_df %>%
  group_by(cluster_kmeans) %>%
  summarise(
    n = n(),

    # Demographics
    age_mean = mean(age_icu_exit),
    age_sd = sd(age_icu_exit),

    # Lactate (back-transformed)
    lactate_median = median(exp(log_lactate)),
    lactate_q1 = quantile(exp(log_lactate), 0.25),
    lactate_q3 = quantile(exp(log_lactate), 0.75),

    # CK (back-transformed)
```

```
    ck_median = median(exp(log_ck)),
    ck_q1 = quantile(exp(log_ck), 0.25),
    ck_q3 = quantile(exp(log_ck), 0.75),

    # Bilirubin (back-transformed)
    bili_median = median(exp(log_bilirubin)),
    bili_q1 = quantile(exp(log_bilirubin), 0.25),
    bili_q3 = quantile(exp(log_bilirubin), 0.75),

    # MAP
    map_mean = mean(mabp_value_num),
    map_sd = sd(mabp_value_num),

    # Outcome
    deaths = sum(icu_dead),
    mortality_pct = mean(icu_dead) * 100,

    .groups = "drop"
  )
```

```
cat("\nDetailed Cluster Profiles:\n")
```

```
##
## Detailed Cluster Profiles:
```

```
print(cluster_profile, digits = 2)
```

```
## # A tibble: 3 x 17
##   cluster_kmeans     n age_mean age_sd lactate_median lactate_q1 lactate_q3
##   <fct>          <int>    <dbl>  <dbl>          <dbl>      <dbl>      <dbl>
## 1 Cluster 1        591     70.3   12.9            1.4        1          2.1
## 2 Cluster 2        314     68.2   13.3            4.1        2.5        6.98
## 3 Cluster 3        239     50.0   15.7            2.8        1.8        4.1
## # i 10 more variables: ck_median <dbl>, ck_q1 <dbl>, ck_q3 <dbl>,
## #   bili_median <dbl>, bili_q1 <dbl>, bili_q3 <dbl>, map_mean <dbl>,
## #   map_sd <dbl>, deaths <dbl>, mortality_pct <dbl>
```

```
# Clinical interpretation
cat("\n\nCLINICAL INTERPRETATION OF CLUSTERS:\n")
```

```
##
##
## CLINICAL INTERPRETATION OF CLUSTERS:
```

```
cat(rep("=", 70), "\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

```r
for (i in 1:optimal_k) {
  cluster_data <- cluster_profile %>% filter(cluster_kmeans == paste0("Cluster ", i))

  cat(sprintf("\n%s (n=%d, mortality %.1f%%):\n",
              paste0("Cluster ", i),
              cluster_data$n,
              cluster_data$mortality_pct))

  cat(sprintf("  Age: %.0f ± %.0f years\n",
              cluster_data$age_mean,
              cluster_data$age_sd))
  cat(sprintf("  Lactate: %.1f [%.1f-%.1f] mmol/L\n",
              cluster_data$lactate_median,
              cluster_data$lactate_q1,
              cluster_data$lactate_q3))
  cat(sprintf("  CK: %.0f [%.0f-%.0f] IU/L\n",
              cluster_data$ck_median,
              cluster_data$ck_q1,
              cluster_data$ck_q3))
  cat(sprintf("  Bilirubin: %.1f [%.1f-%.1f] mg/dL\n",
              cluster_data$bili_median,
              cluster_data$bili_q1,
              cluster_data$bili_q3))
  cat(sprintf("  MAP: %.0f ± %.0f mmHg\n",
              cluster_data$map_mean,
              cluster_data$map_sd))

  # Interpret cluster
  if (cluster_data$lactate_median < 2 && cluster_data$mortality_pct < 30) {
    cat("  INTERPRETATION: LOW RISK - Relatively stable patients\n")
  } else if (cluster_data$lactate_median >= 4 || cluster_data$mortality_pct > 60) {
    cat("  INTERPRETATION: HIGH RISK - Severely ill, high mortality\n")
  } else {
    cat("  INTERPRETATION: MODERATE RISK - Mixed severity\n")
  }
}
```

```
##
## Cluster 1 (n=591, mortality 20.8%):
##   Age: 70 ± 13 years
##   Lactate: 1.4 [1.0-2.1] mmol/L
##   CK: 99 [55-217] IU/L
##   Bilirubin: 0.6 [0.4-0.9] mg/dL
##   MAP: 82 ± 19 mmHg
##   INTERPRETATION: LOW RISK - Relatively stable patients
##
## Cluster 2 (n=314, mortality 48.7%):
##   Age: 68 ± 13 years
##   Lactate: 4.1 [2.5-7.0] mmol/L
##   CK: 217 [88-597] IU/L
##   Bilirubin: 2.5 [1.3-5.2] mg/dL
##   MAP: 74 ± 18 mmHg
##   INTERPRETATION: HIGH RISK - Severely ill, high mortality
##
```

```
## Cluster 3 (n=239, mortality 25.1%):
##   Age: 50 ± 16 years
##   Lactate: 2.8 [1.8-4.1] mmol/L
##   CK: 1465 [528-4286] IU/L
##   Bilirubin: 0.5 [0.4-0.9] mg/dL
##   MAP: 86 ± 20 mmHg
##   INTERPRETATION: MODERATE RISK - Mixed severity
```

The cluster profiling reveals three clinically distinct ICU patient phenotypes with different risk profiles and pathophysiological characteristics. Cluster 1 (n=591, 20.8% mortality) represents a “low-risk elderly stable” group with older patients (70 years) who have relatively normal lactate (1.4 mmol/L), low creatine kinase, stable hemodynamics (MAP 82 mmHg), and preserved liver function, suggesting age-related admissions with less acute physiological derangement. Cluster 2 (n=314, 48.7% mortality) identifies a “high-risk multi-organ dysfunction” phenotype characterized by severe hyperlactatemia (4.1 mmol/L), significantly elevated bilirubin (2.5 mg/dL), hemodynamic instability (MAP 74 mmHg), and elevated creatine kinase, indicating patients with profound metabolic derangement, hepatic dysfunction, and tissue hypoperfusion who face nearly 50% mortality risk. Cluster 3 (n=239, 25.1% mortality) captures a “moderate-risk younger trauma/rhabdomyolysis” group with substantially younger patients (50 years), extremely elevated creatine kinase (1,465 IU/L suggesting muscle injury), moderately elevated lactate, preserved blood pressure, and normal bilirubin, likely representing trauma, seizure, or rhabdomyolysis cases with intermediate prognosis despite marked CK elevation, demonstrating that younger age and single-organ pathology confer better outcomes than multi-organ failure.

```r
# Save clustering results
clustering_results <- list(
  optimal_k = optimal_k,
  kmeans_model = kmeans_result,
  hc_model = hc_ward,
  gmm_model = gmm_result,
  mortality_by_cluster = mortality_by_cluster,
  cluster_profiles = cluster_profile,
  patient_assignments = clustering_df
)

if (!dir.exists("../results")) {
dir.create("../results", recursive = TRUE)
}
save(clustering_results, file = "../results/clustering_results.RData")

cat("\nKey findings:\n")
```

```
##
## Key findings:
```

```r
cat(sprintf("  - Identified %d distinct patient clusters\n", optimal_k))
```

```
##   - Identified 3 distinct patient clusters
```

```r
cat(sprintf("  - Mortality ranges from %.1f%% to %.1f%% across clusters\n",
            min(mortality_by_cluster$mortality_rate),
            max(mortality_by_cluster$mortality_rate)))
```

```
##   - Mortality ranges from 20.8% to 48.7% across clusters
```

```r
cat("  - Clusters reflect different severity profiles\n")
```

```
##   - Clusters reflect different severity profiles
```

```r
cat("\nResults saved to: results/clustering_results.RData\n")
```

```
##
## Results saved to: results/clustering_results.RData
```

4.5 ML models (C5.0, Random Forest, LGBoost) + SHAP explanations

```r
# Prepare ML dataset with median imputation
ml_data <- analysis_df %>%
  select(icustay_id, age_icu_exit, gender, icu_dead,
         log_lactate, log_ck, log_bilirubin, mabp_value_num) %>%
  mutate(
    # Impute missing values with median
    log_ck = if_else(is.na(log_ck), median(log_ck, na.rm = TRUE), log_ck),
    log_bilirubin = if_else(is.na(log_bilirubin),
                            median(log_bilirubin, na.rm = TRUE),
                            log_bilirubin),
    mabp_value_num = if_else(is.na(mabp_value_num),
                             median(mabp_value_num, na.rm = TRUE),
                             mabp_value_num),
    # Convert gender to numeric
    gender_num = ifelse(gender == "M", 1, 0),
    # Convert outcome to factor for classification
    icu_dead_f = factor(icu_dead, levels = c(0, 1),
                        labels = c("Survived", "Died"))
  ) %>%
  select(-gender, -icustay_id)

cat(sprintf("Dataset size: %d patients\n", nrow(ml_data)))
```

```
## Dataset size: 22302 patients
```

```r
cat(sprintf("Features: %d\n", ncol(ml_data) - 2))  # Exclude outcomes
```

```
## Features: 6
```

```r
cat(sprintf("Class balance: %.1f%% died, %.1f%% survived\n",
            mean(ml_data$icu_dead) * 100,
            mean(ml_data$icu_dead == 0) * 100))
```

```
## Class balance: 16.0% died, 84.0% survived
```

```
# Train-test split (75-25)
set.seed(42)
train_idx <- createDataPartition(ml_data$icu_dead_f, p = 0.75, list = FALSE)
train_data <- ml_data[train_idx, ]
test_data <- ml_data[-train_idx, ]

cat(sprintf("\nTrain set: %d (%.1f%% died)\n",
            nrow(train_data),
            mean(train_data$icu_dead) * 100))
```

```
##
## Train set: 16727 (16.0% died)
```

```
cat(sprintf("Test set: %d (%.1f%% died)\n",
            nrow(test_data),
            mean(test_data$icu_dead) * 100))
```

```
## Test set: 5575 (16.0% died)
```

Next step is creating a machine learning baseline model, the Logistic regression.

```
# Train control
train_control <- trainControl(
  method = "cv",
  number = 10,
  classProbs = TRUE,
  summaryFunction = twoClassSummary,
  savePredictions = "final"
)

# Train logistic regression
set.seed(42)
model_glm <- train(
  icu_dead_f ~ age_icu_exit + gender_num + log_lactate +
    log_ck + log_bilirubin + mabp_value_num,
  data = train_data,
  method = "glm",
  family = "binomial",
  trControl = train_control,
  metric = "ROC"
)

# Predictions
pred_glm <- predict(model_glm, test_data, type = "prob")[, "Died"]
roc_glm <- roc(test_data$icu_dead_f, pred_glm)

cat(sprintf("Logistic Regression AUC: %.3f (%.3f-%.3f)\n",
            auc(roc_glm), ci.auc(roc_glm)[1], ci.auc(roc_glm)[3]))
```

```
## Logistic Regression AUC: 0.694 (0.674-0.714)
```

The first results as baseline is Area Under Curve around 70%.

4.5.1 Decision Tree: C5.0 ALGORITHM

```
# Train C5.0 decision tree
set.seed(42)
model_c50 <- C5.0(
  icu_dead_f ~ age_icu_exit + gender_num + log_lactate +
    log_ck + log_bilirubin + mabp_value_num,
  data = train_data,
  rules = FALSE,  # Tree format
  trials = 1      # Single tree (no boosting)
)

# Print tree summary
cat("\nC5.0 Tree Summary:\n")
```

```
##
## C5.0 Tree Summary:
```

```
summary(model_c50)
```

```
##
## Call:
## C5.0.formula(formula = icu_dead_f ~ age_icu_exit + gender_num + log_lactate
##  + log_ck + log_bilirubin + mabp_value_num, data = train_data, rules =
##  FALSE, trials = 1)
##
##
## C5.0 [Release 2.07 GPL Edition]      Fri Jan 23 10:00:27 2026
## -------------------------------
##
## Class specified by attribute `outcome'
##
## Read 16727 cases (7 attributes) from undefined.data
##
## Decision tree:
##
## log_lactate <= 1.629241:
## :...log_bilirubin <= 1.93297: Survived (15090/1918)
## :   log_bilirubin > 1.93297:
## :   :...log_lactate <= 1.360977: Survived (325/107)
## :       log_lactate > 1.360977: Died (50/20)
## log_lactate > 1.629241:
## :...log_lactate > 2.261763: Died (280/80)
##     log_lactate <= 2.261763:
##     :...age_icu_exit <= 76: Survived (713/262)
##         age_icu_exit > 76: Died (269/113)
##
##
## Evaluation on training data (16727 cases):
##
##      Decision Tree
##    ----------------
```

```
##    Size      Errors
##
##       6 2500(14.9%)   <<
##
##
##     (a)   (b)    <-classified as
##    ----  ----
##   13841   213    (a): class Survived
##    2287   386    (b): class Died
##
##
##  Attribute usage:
##
##  100.00% log_lactate
##   92.46% log_bilirubin
##    5.87% age_icu_exit
##
##
## Time: 0.0 secs
```

```r
# Extract rules
cat("\n\nDECISION RULES FROM C5.0:\n")
```

```
##
##
## DECISION RULES FROM C5.0:
```

```r
cat(rep("=", 70), "\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

```r
model_c50_rules <- C5.0(
  icu_dead_f ~ age_icu_exit + gender_num + log_lactate +
    log_ck + log_bilirubin + mabp_value_num,
  data = train_data,
  rules = TRUE,  # Rule format
  trials = 1
)

cat("\n")
```

```r
summary(model_c50_rules)
```

```
##
## Call:
## C5.0.formula(formula = icu_dead_f ~ age_icu_exit + gender_num + log_lactate
##  + log_ck + log_bilirubin + mabp_value_num, data = train_data, rules =
##  TRUE, trials = 1)
##
##
## C5.0 [Release 2.07 GPL Edition]      Fri Jan 23 10:00:27 2026
## -------------------------------
```

```
##
## Class specified by attribute `outcome'
##
## Read 16727 cases (7 attributes) from undefined.data
##
## Rules:
##
## Rule 1: (11902/1458, lift 1.0)
##  age_icu_exit <= 76
##  log_lactate <= 2.261763
##  ->  class Survived  [0.877]
##
## Rule 2: (14540/1840, lift 1.0)
##  log_lactate <= 1.360977
##  ->  class Survived  [0.873]
##
## Rule 3: (15090/1918, lift 1.0)
##  log_lactate <= 1.629241
##  log_bilirubin <= 1.93297
##  ->  class Survived  [0.873]
##
## Rule 4: (280/80, lift 4.5)
##  log_lactate > 2.261763
##  ->  class Died  [0.713]
##
## Rule 5: (336/124, lift 3.9)
##  age_icu_exit > 76
##  log_lactate > 1.629241
##  ->  class Died  [0.630]
##
## Rule 6: (140/56, lift 3.7)
##  log_lactate > 1.360977
##  log_bilirubin > 1.93297
##  ->  class Died  [0.599]
##
## Default class: Survived
##
##
## Evaluation on training data (16727 cases):
##
##          Rules
##    ----------------
##      No      Errors
##
##       6 2504(15.0%)   <<
##
##
##     (a)   (b)    <-classified as
##    ----  ----
##   13861   193    (a): class Survived
##    2311   362    (b): class Died
##
##
##  Attribute usage:
```

```
##
##  100.00% log_lactate
##   91.05% log_bilirubin
##   73.16% age_icu_exit
##
##
## Time: 0.0 secs
```

```r
# Predictions
pred_c50 <- predict(model_c50, test_data, type = "prob")[, "Died"]
roc_c50 <- roc(test_data$icu_dead_f, pred_c50)

cat(sprintf("\nC5.0 Decision Tree AUC: %.3f (%.3f-%.3f)\n",
            auc(roc_c50), ci.auc(roc_c50)[1], ci.auc(roc_c50)[3]))
```

```
##
## C5.0 Decision Tree AUC: 0.600 (0.585-0.615)
```

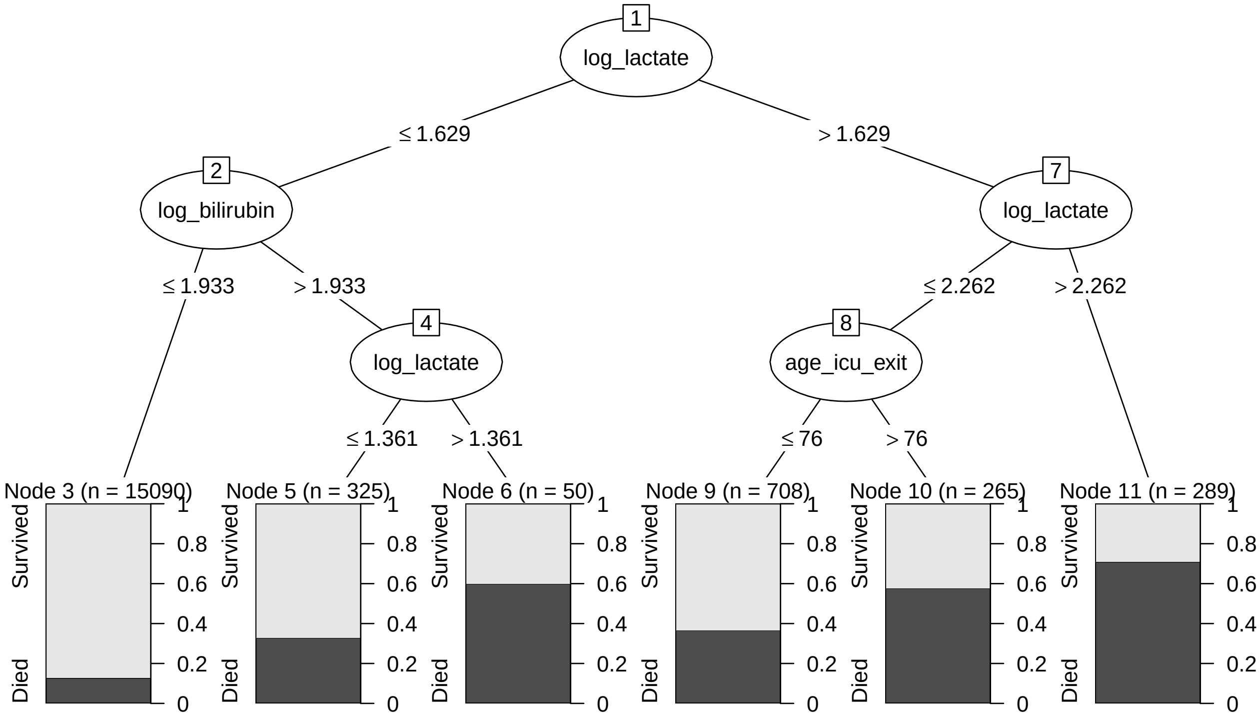
```r
# Variable importance
c50_importance <- C5imp(model_c50, metric = "usage")
cat("\nC5.0 Variable Importance (usage-based):\n")
```

```
##
## C5.0 Variable Importance (usage-based):
```

```r
print(c50_importance)
```

```
##                Overall
## log_lactate     100.00
## log_bilirubin    92.46
## age_icu_exit      5.87
## gender_num        0.00
## log_ck            0.00
## mabp_value_num    0.00
```

```r
# Show in notebook
plot(model_c50)
```




```r
# Save to file
png(
  "../figures/figure14_c50_model_plot.png",
  width = 8,
  height = 6,
  units = "in",
  res = 300
)
plot(model_c50)
dev.off()
```

```
## pdf
##   2
```

```r
summary(model_c50_rules)
```

```
##
## Call:
## C5.0.formula(formula = icu_dead_f ~ age_icu_exit + gender_num + log_lactate
##  + log_ck + log_bilirubin + mabp_value_num, data = train_data, rules =
##  TRUE, trials = 1)
##
##
## C5.0 [Release 2.07 GPL Edition]      Fri Jan 23 10:00:27 2026
## -------------------------------
##
## Class specified by attribute `outcome'
##
## Read 16727 cases (7 attributes) from undefined.data
##
```

```
## Rules:
##
## Rule 1: (11902/1458, lift 1.0)
##  age_icu_exit <= 76
##  log_lactate <= 2.261763
##  ->  class Survived  [0.877]
##
## Rule 2: (14540/1840, lift 1.0)
##  log_lactate <= 1.360977
##  ->  class Survived  [0.873]
##
## Rule 3: (15090/1918, lift 1.0)
##  log_lactate <= 1.629241
##  log_bilirubin <= 1.93297
##  ->  class Survived  [0.873]
##
## Rule 4: (280/80, lift 4.5)
##  log_lactate > 2.261763
##  ->  class Died  [0.713]
##
## Rule 5: (336/124, lift 3.9)
##  age_icu_exit > 76
##  log_lactate > 1.629241
##  ->  class Died  [0.630]
##
## Rule 6: (140/56, lift 3.7)
##  log_lactate > 1.360977
##  log_bilirubin > 1.93297
##  ->  class Died  [0.599]
##
## Default class: Survived
##
##
## Evaluation on training data (16727 cases):
##
##          Rules
##    ----------------
##      No      Errors
##
##       6 2504(15.0%)   <<
##
##
##     (a)   (b)    <-classified as
##    ----  ----
##   13861   193    (a): class Survived
##    2311   362    (b): class Died
##
##
##  Attribute usage:
##
##  100.00% log_lactate
##   91.05% log_bilirubin
##   73.16% age_icu_exit
##
```

```
##
## Time: 0.0 secs
```

The C5.0 decision tree generated six interpretable rules that classify ICU patients into survival or death outcomes based on lactate, age, and bilirubin levels. Rules 1-3 identify survival patterns: Rule 1 predicts survival for younger patients (76 years) with moderate lactate levels (2.26 log mmol/L or ~9.6 mmol/L), covering 11,902 cases with 87.7% accuracy; Rule 2 predicts survival for all patients with low lactate (1.36 log mmol/L or ~3.9 mmol/L), the most protective factor covering 14,540 cases with 87.3% accuracy; and Rule 3 combines moderate lactate (1.63 log mmol/L or ~5.1 mmol/L) with normal bilirubin (1.93 log mg/dL or ~6.9 mg/dL) to predict survival in 15,090 cases with 87.3% accuracy.

Rules 4-6 identify high-risk mortality patterns: Rule 4 shows that severe hyperlactatemia (>2.26 log mmol/L or >9.6 mmol/L) alone predicts death with 71.3% accuracy and a 4.5-fold lift; Rule 5 identifies elderly patients (>76 years) with elevated lactate (>1.63 log mmol/L or >5.1 mmol/L) as high-risk with 63.0% accuracy and 3.9-fold lift; and Rule 6 reveals that combined elevated lactate (>1.36 log mmol/L or >3.9 mmol/L) with high bilirubin (>1.93 log mg/dL or >6.9 mg/dL) predicts death with 59.9% accuracy and 3.7-fold lift, indicating multi-organ dysfunction. The model achieves 85% overall accuracy on training data, with lactate being the most critical variable (used 100% of the time), followed by bilirubin (91%) and age (73%), demonstrating that lactate thresholds around 4-10 mmol/L are key decision points, with elderly age and hepatic dysfunction significantly modifying mortality risk.

4.5.2 Random Forest

```r
# Train Random Forest
set.seed(42)
model_rf <- randomForest(
  icu_dead_f ~ age_icu_exit + gender_num + log_lactate +
    log_ck + log_bilirubin + mabp_value_num,
  data = train_data,
  ntree = 500,
  mtry = 3,  # sqrt(6)  2.5
  importance = TRUE,
  nodesize = 5
)

cat("\nRandom Forest Model:\n")
```

```
##
## Random Forest Model:
```

```r
print(model_rf)
```

```
##
## Call:
##  randomForest(formula = icu_dead_f ~ age_icu_exit + gender_num +      log_lactate + log_ck + log_bili
##                Type of random forest: classification
##                      Number of trees: 500
## No. of variables tried at each split: 3
##
```

```
##         OOB estimate of  error rate: 15.67%
## Confusion matrix:
##          Survived Died class.error
## Survived    13641  413  0.02938665
## Died         2208  465  0.82603816
```

```
# Predictions
pred_rf <- predict(model_rf, test_data, type = "prob")[, "Died"]
roc_rf <- roc(test_data$icu_dead_f, pred_rf)

cat(sprintf("\nRandom Forest AUC: %.3f (%.3f-%.3f)\n",
            auc(roc_rf), ci.auc(roc_rf)[1], ci.auc(roc_rf)[3]))
```

```
##
## Random Forest AUC: 0.716 (0.697-0.735)
```

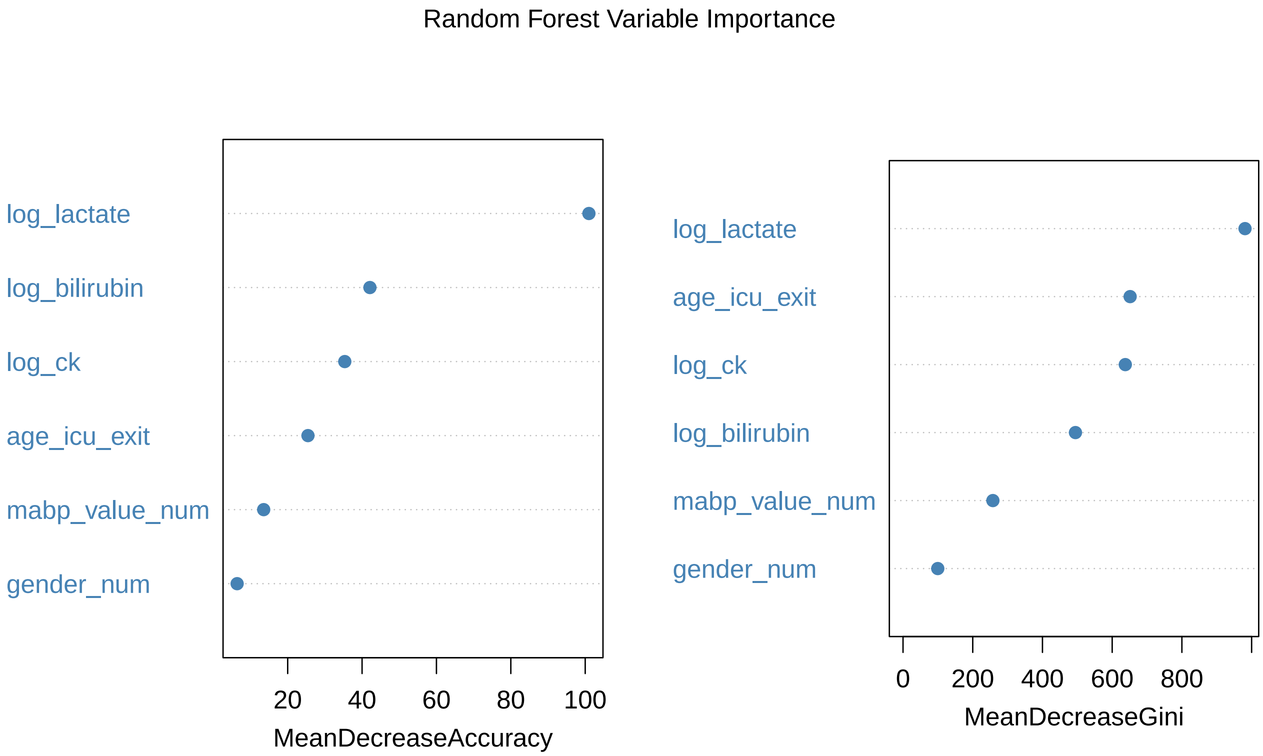
```
# Variable importance
cat("\nRandom Forest Variable Importance:\n")
```

```
##
## Random Forest Variable Importance:
```

```
rf_importance <- importance(model_rf)
print(rf_importance)
```

```
##                 Survived      Died MeanDecreaseAccuracy MeanDecreaseGini
## age_icu_exit   15.424546 24.013769            25.450912        651.49941
## gender_num      7.947188 -1.593097             6.404013         99.63562
## log_lactate    65.254524 99.078168           100.993437        981.06751
## log_ck         21.255167 37.192698            35.350402        637.61971
## log_bilirubin  30.044527 35.203375            42.109078        494.65520
## mabp_value_num 14.413168  1.523863            13.545550        257.75092
```

```
# Plot importance
varImpPlot(model_rf,
           main = "Random Forest Variable Importance",
           pch = 19,
           col = "steelblue",
           cex = 1.2)
```




```r
png("../figures/figure15_rf_importance.png",
    width = 10, height = 7, units = "in", res = 300)

varImpPlot(model_rf,
           main = "Random Forest Variable Importance",
           pch = 19,
           col = "steelblue",
           cex = 1.2)

dev.off()
```

```
## pdf
##   2
```

4.5.3 LightGBM Gradient Boosting

```r
# Train LightGBM

train_matrix <- model.matrix(icu_dead_f ~ . - icu_dead - 1, data = train_data)
test_matrix <- model.matrix(icu_dead_f ~ . - icu_dead - 1, data = test_data)

train_labels <- as.numeric(train_data$icu_dead_f == "Died")
test_labels <- as.numeric(test_data$icu_dead_f == "Died")

# dtest <- lgb.DMatrix(data = test_matrix, label = test_labels)

dtrain <- lgb.Dataset(
```

```
  data = train_matrix,
  label = train_labels
)

dtrain <- lgb.Dataset(
  data = test_matrix,
  label = test_labels
)

# LGBoost parameters
params <- list(
  objective = "binary",
  metric = "auc",
  learning_rate = 0.05,
  num_leaves = 15,      # similar complexity to C5.0
  max_depth = 4,
  min_data_in_leaf = 20,
  feature_fraction = 0.9,
  bagging_fraction = 0.8,
  bagging_freq = 5
)


# Training
set.seed(42)

model_lgb <- lgb.train(
  params = params,
  data = dtrain,
  nrounds = 300,
  verbose = -1
)


# Predictions
pred_lgb <- predict(model_lgb, test_matrix)

roc_obj <- pROC::roc(test_labels, pred_lgb)
cat(sprintf("Test AUC: %.3f\n", pROC::auc(roc_obj)))
```

```
## Test AUC: 0.840
```

```
# Feature importance
importance_lgb <- lgb.importance(model = model_lgb)
cat("\nLightGBoost Feature Importance:\n")
```

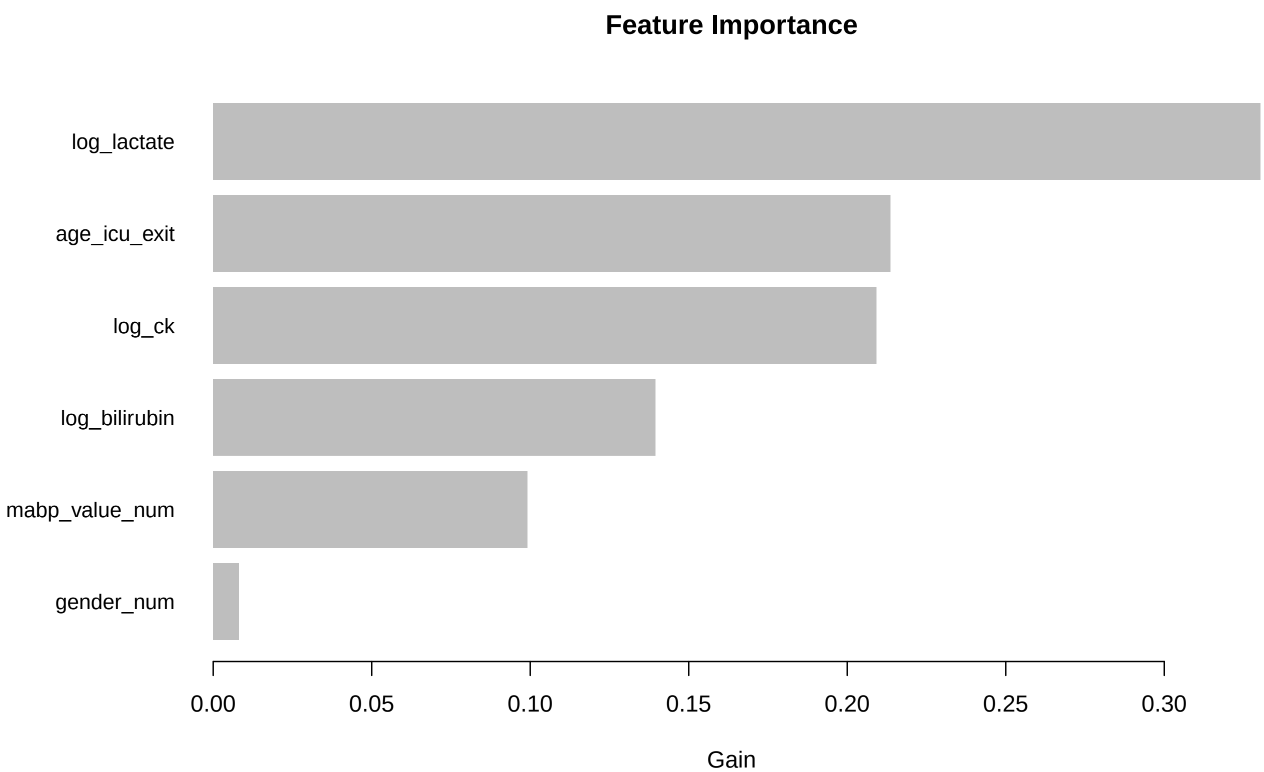
```
##
## LightGBoost Feature Importance:
```

```
print(importance_lgb)
```

```
##            Feature       Gain      Cover  Frequency
```

```
##              <char>      <num>      <num>      <num>
## 1:     log_lactate 0.33037191 0.21216104 0.22376439
## 2:    age_icu_exit 0.21368324 0.17612753 0.22545701
## 3:          log_ck 0.20917105 0.23959126 0.23967502
## 4:   log_bilirubin 0.13953708 0.16066621 0.15165877
## 5: mabp_value_num 0.09915896 0.20240642 0.14353419
## 6:      gender_num 0.00807776 0.00904754 0.01591063
```

```r
# Plot importance
lgb.plot.importance(importance_lgb)
```




```r
png("../figures/figure16_lgb_importance.png",
    width = 10, height = 7, units = "in", res = 300)
lgb.plot.importance(importance_lgb)
dev.off()
```

```
## pdf
##   2
```

4.5.4 Model Comparison

```r
# Create comparison table
model_comparison <- data.frame(
  Model = c("Logistic Regression", "C5.0 Decision Tree",
            "Random Forest", "LightGBoost"),
  AUC = c(auc(roc_glm), auc(roc_c50), auc(roc_rf), auc(roc_obj)),
```

```
  CI_lower = c(ci.auc(roc_glm)[1], ci.auc(roc_c50)[1],
               ci.auc(roc_rf)[1], ci.auc(roc_obj)[1]),
  CI_upper = c(ci.auc(roc_glm)[3], ci.auc(roc_c50)[3],
               ci.auc(roc_rf)[3], ci.auc(roc_obj)[3])
) %>%
  arrange(desc(AUC)) %>%
  mutate(
    AUC_CI = sprintf("%.3f (%.3f-%.3f)", AUC, CI_lower, CI_upper)
  )

cat("\nModel Performance Comparison:\n")
```

```
##
## Model Performance Comparison:
```

```
print(model_comparison %>% select(Model, AUC_CI), row.names = FALSE)
```

```
##                Model              AUC_CI
##          LightGBoost 0.840 (0.825-0.854)
##        Random Forest 0.716 (0.697-0.735)
##  Logistic Regression 0.694 (0.674-0.714)
##   C5.0 Decision Tree 0.600 (0.585-0.615)
```

```
# Plot ROC curves
png("../figures/figure17_ml_roc_comparison.png",
    width = 10, height = 10, units = "in", res = 300)

plot(roc_glm, col = "blue", lwd = 2.5, legacy.axes = TRUE,
     main = "ROC Curves: Model Comparison",
     cex.main = 1.5, cex.lab = 1.3)
plot(roc_c50, col = "green", lwd = 2.5, add = TRUE)
plot(roc_rf, col = "purple", lwd = 2.5, add = TRUE)
plot(roc_obj, col = "yellow", lwd = 2.5, add = TRUE)

abline(a = 0, b = 1, lty = 2, col = "gray50", lwd = 1.5)

legend("bottomright",
       legend = c(
         sprintf("Logistic Regression (AUC = %.3f)", auc(roc_glm)),
         sprintf("C5.0 Decision Tree (AUC = %.3f)", auc(roc_c50)),
         sprintf("Random Forest (AUC = %.3f)", auc(roc_rf)),
          sprintf("LightGBoost (AUC = %.3f)", auc(roc_obj))
       ),
       col = c("blue", "green", "purple", "yellow"),
       lwd = 2.5,
       cex = 1.1,
       bty = "n")

grid(col = "gray", lty = "dotted")

# Save comparison table
write_csv(model_comparison, "../results/ml_model_comparison.csv")
```

4.5.5 SHAP Values Analysis (Model Explainability)

```
# Prepare data for SHAP
test_data_matrix <- test_data %>%
  select(age_icu_exit, gender_num, log_lactate,
         log_ck, log_bilirubin, mabp_value_num) %>%
  as.matrix()

train_data_matrix <- train_data %>%
  select(age_icu_exit, gender_num, log_lactate,
         log_ck, log_bilirubin, mabp_value_num) %>%
  as.matrix()

# Calculate SHAP values for LightGBoost
cat("\nCalculating SHAP values for LightGBoost model...\n")
```

```
##
## Calculating SHAP values for LightGBoost model...
```

```
shap_lgb <- shapviz(
  model_lgb,
  X_pred = test_data_matrix
)
```

```
# Summary plot
png("../figures/figure18_shap_summary.png",
    width = 10, height = 8, units = "in", res = 300)

sv_importance(shap_lgb, kind = "beeswarm")

# Bar plot of mean absolute SHAP values
png("../figures/figure19_shap_importance_bar.png",
    width = 10, height = 7, units = "in", res = 300)

sv_importance(shap_lgb, kind = "bar")

# Dependence plots for top features
png("../figures/figure20_shap_dependence_lactate.png",
    width = 10, height = 7, units = "in", res = 300)

sv_dependence(shap_lgb, v = "log_lactate", color_var = "auto")

# Get SHAP values as dataframe
shap_values_df <- as.data.frame(shap_lgb$S)
colnames(shap_values_df) <- c("age", "gender", "lactate", "ck", "bilirubin", "mabp")

# Calculate mean absolute SHAP values
shap_importance <- colMeans(abs(shap_values_df)) %>%
  sort(decreasing = TRUE)

cat("\nMean Absolute SHAP Values (Feature Importance):\n")
```

```
##
## Mean Absolute SHAP Values (Feature Importance):
```

```r
print(round(shap_importance, 4))
```

```
##   lactate        ck bilirubin    gender       age      mabp
##    1.4232    1.1359    0.6321    0.2976    0.1627    0.0517
```

```r
# SHAP interpretation
cat("\n\nSHAP INTERPRETATION:\n")
```

```
##
##
## SHAP INTERPRETATION:
```

```r
cat(rep("=", 70), "\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

```r
cat("SHAP values measure each feature's contribution to individual predictions.\n")
```

```
## SHAP values measure each feature's contribution to individual predictions.
```

```r
cat("Higher |SHAP| = greater impact on mortality prediction.\n\n")
```

```
## Higher |SHAP| = greater impact on mortality prediction.
```

```r
cat("Key insights:\n")
```

```
## Key insights:
```

```r
cat(sprintf("1. %s has the highest average impact (%.4f)\n",
            names(shap_importance)[1], shap_importance[1]))
```

```
## 1. lactate has the highest average impact (1.4232)
```

```r
cat(sprintf("2. %s is second most important (%.4f)\n",
            names(shap_importance)[2], shap_importance[2]))
```

```
## 2. ck is second most important (1.1359)
```

```r
cat(sprintf("3. %s has the least impact (%.4f)\n",
            names(shap_importance)[length(shap_importance)],
            shap_importance[length(shap_importance)]))
```

```
## 3. mabp has the least impact (0.0517)
```

```r
cat("\nConclusion: ", names(shap_importance)[1],
    " is the most critical predictor across all patients.\n")
```

```
##
## Conclusion:  lactate  is the most critical predictor across all patients.
```

4.6 Feature importance comparison across models

```r
# Combine importance from all models
importance_combined <- data.frame(
  Feature = c("age_icu_exit", "gender_num", "log_lactate",
              "log_ck", "log_bilirubin", "mabp_value_num"),
  C50 = c50_importance[,],
  RandomForest = rf_importance[, "MeanDecreaseGini"],
  LGBoost = importance_lgb$Gain[match(
    c("age_icu_exit", "gender_num", "log_lactate",
      "log_ck", "log_bilirubin", "mabp_value_num"),
    importance_lgb$Feature
  )],
  SHAP = shap_importance[c("age", "gender", "lactate", "ck", "bilirubin", "mabp")]
)

# Normalize to 0-100
importance_normalized <- importance_combined %>%
  mutate(
    C50 = C50 / max(C50, na.rm = TRUE) * 100,
    RandomForest = RandomForest / max(RandomForest) * 100,
    LGBoost = LGBoost / max(LGBoost) * 100,
    SHAP = SHAP / max(SHAP) * 100
  )

cat("\nNormalized Feature Importance (0-100):\n")
```

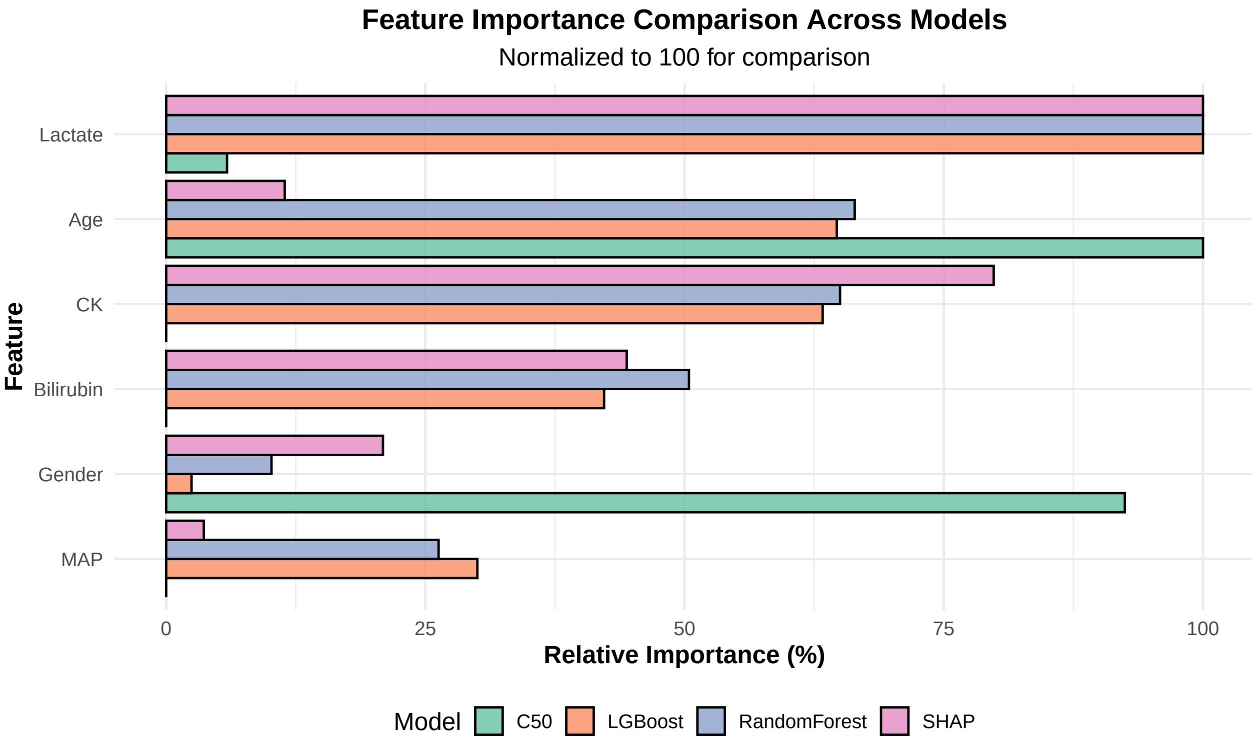
```
##
## Normalized Feature Importance (0-100):
```

```r
importance_normalized %>%
  select(-Feature) %>%
  round(1)
```

```
##                  C50 RandomForest LGBoost  SHAP
## age_icu_exit   100.0         66.4    64.7  11.4
## gender_num      92.5         10.2     2.4  20.9
## log_lactate      5.9        100.0   100.0 100.0
## log_ck           0.0         65.0    63.3  79.8
## log_bilirubin    0.0         50.4    42.2  44.4
## mabp_value_num   0.0         26.3    30.0   3.6
```

```r
# Save importance comparison
write_csv(importance_normalized, "../results/feature_importance_comparison.csv")

# Plot comparison
png("../figures/figure21_importance_comparison.png",
    width = 12, height = 8, units = "in", res = 300)

importance_long <- importance_normalized %>%
  pivot_longer(cols = c(C50, RandomForest, LGBoost, SHAP),
               names_to = "Model",
               values_to = "Importance") %>%
  mutate(
    Feature = case_when(
      Feature == "age_icu_exit" ~ "Age",
      Feature == "gender_num" ~ "Gender",
      Feature == "log_lactate" ~ "Lactate",
      Feature == "log_ck" ~ "CK",
      Feature == "log_bilirubin" ~ "Bilirubin",
      Feature == "mabp_value_num" ~ "MAP"
    )
  )
```

```r
library(ggplot2)
fig21 <- ggplot(importance_long, aes(x = reorder(Feature, Importance),
                              y = Importance, fill = Model)) +
  geom_col(position = "dodge", alpha = 0.8, color = "black") +
  coord_flip() +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Feature Importance Comparison Across Models",
    subtitle = "Normalized to 100 for comparison",
    x = "Feature",
    y = "Relative Importance (%)",
    fill = "Model"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    axis.title = element_text(face = "bold"),
    legend.position = "bottom"
  )
fig21
```



4.6.1 Save the results

```
ml_results <- list(
  model_glm = model_glm,
  model_c50 = model_c50,
  model_c50_rules = model_c50_rules,
  model_rf = model_rf,
  model_lgb = model_lgb,
  model_comparison = model_comparison,
  importance_comparison = importance_normalized,
  shap_values = shap_values_df,
  test_predictions = data.frame(
    actual = test_data$icu_dead_f,
    pred_glm = pred_glm,
    pred_c50 = pred_c50,
    pred_rf = pred_rf,
    pred_lgb = pred_lgb
  )
)

save(ml_results, file = "../results/ml_complete_results.RData")
```

5 Next steps

6 Conclusions

6.1 Final thoughts

7 References